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(54) **POLYHYDROXYALKANOATE PRODUCTION AND RELATED PROCESSES**

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(57) **ABSTRACT**

Embodiments of the invention relate generally to processes for the production and processing of polyhydroxyalkanoates (PHA) from carbon sources. In several embodiments, PHAs are produced at high efficiencies from carbon-containing gases through the utilization of a regenerative polymerization system.

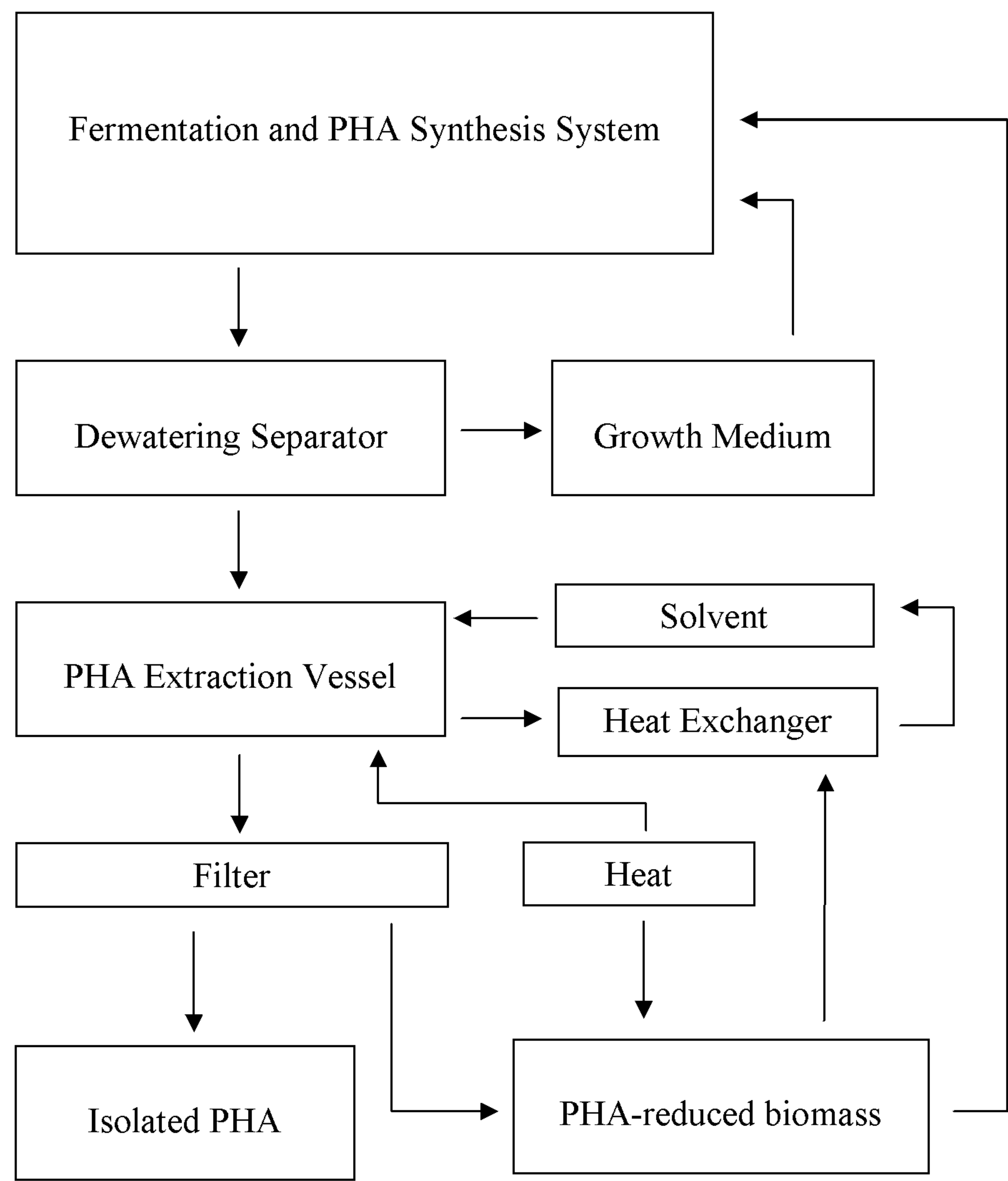


FIG. 1

POLYHYDROXYALKANOATE PRODUCTION AND RELATED PROCESSES

RELATED APPLICATIONS

[0001] This application is a continuation of co-pending U.S. patent application Ser. No. 13/392,502 filed Feb. 2, 2012, which is the U.S. National Phase of International Application No. PCT/US2010/047052, filed Aug. 27, 2010, which claims the benefit of U.S. Provisional Application Nos. 61/237,606, 61/237,609, 61/237,635, 61/237,603, 61/237,616, 61/237,615, 61/237,620, 61/237,643, 61/237,633, 61/237,630, 61/237,626, 61/237,642, 61/237,639, and 61/237,627, all filed on Aug. 27, 2009, the entire disclosure of each application in the priority chain is incorporated by reference herein.

BACKGROUND

Field of the Invention

[0002] Embodiments of the invention relate to an improved process for the production and processing of polyhydroxyalkanoates, and specifically to a process for the production of polyhydroxyalkanoates from carbon-containing gases.

Description of the Related Art

[0003] Polyhydroxyalkanoates (PHAs) are thermoplastic polyesters that serve as energy storage vehicles in microorganisms. PHAs are biodegradable in both aerobic and anaerobic conditions, are biocompatible with mammalian tissues, and, as thermoplastics, can be used as alternatives to fossil fuel-based plastics such as polypropylene, polyethylene, and polystyrene. In comparison to petrochemical-based plastics, which are neither biodegradable nor made from sustainable sources of carbon, PHA plastics afford significant environmental benefits.

[0004] The utilization of food crop derived sugars in genetically engineered microorganism-based aqueous fermentation systems is often regarded as the most efficient and economical platform for PHA production. Specifically, sugar-based PHA production processes are capable of generating high density fermentation cultures and high PHA inclusion concentrations, and, by maximizing the cell culture density and PHA inclusion concentration therein, it is believed that carbon, chemical, and energy efficiencies are also maximized. For example, comparing a low cell and PHA concentration process to a high cell and PHA concentration process, a low concentration process requires significantly more, per given unit of PHA-containing biomass, i) energy for dewatering cells prior to PHA extraction treatment, i) liquid culture volume, and associated chemicals, mixing energy, and heat removal energy, and iii) both energy and chemicals for separating PHA from biomass. Accordingly, whereas the sugar-based genetically-engineered microorganism PHA process yields maximized cell densities and PHA concentrations relative to low concentration processes, it is also regarded as the most carbon, chemical, energy, and, thus, cost efficient PHA production method.

[0005] Unfortunately, despite these maximized efficiency advantages, sugar-based PHA production remains many times more expensive than fossil fuel-based plastics production. Thus, given the apparent efficiency maximization of the high density sugar-derived PHA production process, PHAs

are widely considered to be fundamentally unable to compete with fossil fuel-based plastics on energy, chemical, and cost efficiency.

SUMMARY

[0006] Despite the environmental advantages of PHAs, the high cost of PHA production relative to the low cost of fossil fuel-based plastics production has significantly limited the industrial production and commercial adoption of PHAs.

[0007] To reduce the carbon input cost of the PHA production process, carbon-containing industrial off-gases, such as carbon dioxide, methane, and volatile organic compounds, have been proposed as an alternative to food crop-based sources of carbon. In addition to the wide availability and low cost of carbon-containing gases, carbon-containing gases also do not present the environmental challenges associated with food crop-derived sources of carbon. Specifically, whereas food crop-based carbon substrates require land, fertilizers, pesticides, and fossil fuels to produce, and also generate greenhouse gas emissions during the course of production, carbon-containing off-gases do not require new inputs of land, fertilizers, pesticides, or fossil fuels to generate. Thus, on both an economic and environmental basis, the utilization of carbon emissions for the production of PHA would appear to offer significant advantages over sugar-based PHA production processes.

[0008] Unfortunately, the fermentation of carbon-containing gases presents technical challenges and stoichiometric limitations that have, in the past, rendered the gas-to-PHA production process significantly more energy and chemical intensive, and thus more costly, than the food crop-based PHA production process. Specifically, these technical challenges and stoichiometric limitations include: low mass transfer rates, low microorganism growth rates, extended polymerization times, low cell densities, high oxygen demand, and low PHA cellular inclusion concentrations. Whereas sugar-based fermentation systems have the ability to generate high cellular densities and PHA inclusion concentrations, based on cell morphology and mass transfer constraints, carbon-containing gas-based fermentation processes typically generate 10-30% of the biomass and intracellular PHA inclusion concentrations achieved in sugar-based processes. As a result, the ratio of energy-to-PHA required to carry out upstream carbon injection, optional oxygen injection, and culture mixing, as well as downstream PHA purification, significantly exceeds the energy-to-PHA ratio required for sugar-based PHA production methods, thereby rendering the emissions-based process uncompetitive when compared to both petroleum-based plastics and sugar-based PHAs.

[0009] In light of the potential environmental advantages and carbon cost efficiencies of utilizing carbon-containing gases as a source of carbon for PHA production, there exists a significant need to reduce the energy, chemical, and carbon input-to-PHA output ratio in a carbon emissions-based PHA production system, and thereby render carbon gas-derived PHA economically competitive with petrochemical-based plastics.

[0010] Thus, in several embodiments, the present invention relates to a novel process for the conversion of carbon-containing gases into PHAs at previously unattainable energy and carbon PHA conversion ratios.

[0011] In some embodiments, the invention also relates to a process that generates a carbon emissions-based PHA

material that is cost-competitive with both food crop-based PHAs and fossil fuel-based thermoplastics.

[0012] While PHAs are widely considered to be noncompetitive with fossil fuel-based plastics on energy, chemical, and cost efficiency, several embodiments of the invention relate to a process for producing PHAs from carbon-containing gases that yields unexpectedly improved energy, carbon, chemical, and cost efficiencies over sugar-based PHA production methodologies.

[0013] More specifically, certain embodiments of the invention provide high efficiency, high density, high PHA concentration processes for the production of PHA from carbon-containing gases, comprising the steps of: (a) providing a microorganism culture comprising PHA-containing biomass, (b) removing a portion of the PHA-containing biomass from the culture, (c) extracting a portion of PHA from the removed culture to produce isolated PHA and PHA-reduced biomass, (d) purifying the isolated PHA, and (e) returning the PHA-reduced biomass to the culture to cause the culture to convert the carbon within the PHA-reduced biomass into PHA. In several embodiments, carbon output from the system is wholly or substantially only in the form of PHA.

[0014] In several embodiments, a system for using a microorganism culture to convert a carbon-containing gas into PHA at high efficiencies is provided. Microorganisms are cultured using a combination of one or more carbon-containing gases and PHA-reduced biomass, or derivatives thereof, as sources of carbon to produce PHA-containing biomass. A portion of the PHA-containing biomass is then removed from the culture, and PHA is extracted from the removed PHA-containing biomass to create substantially PHA-reduced biomass and substantially isolated PHA.

[0015] Typically, PHA is present in the PHA-containing biomass of gas-utilizing microorganisms at concentrations in the range of about 5%-60%, and approximately 40-95% of the PHA-containing biomass is discarded from the system following PHA extraction. In some cases, PHA is present in gas-utilization microorganisms in the range of about 1-90%, including at about 1%, 3%, 5%, 7%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 60%, 70%, 80%, or 90%, and approximately 10-99% of the PHA-containing biomass is discarded from the system following PHA extraction, including 99%, 97%, 95%, 93%, 90%, 85%, 80%, 75%, 70%, 65%, 60%, 55%, 50%, 40%, 30%, 20%, or 10% of the PHA-containing biomass. Rather than discarding the remaining, e.g., 40-95% of the PHA-reduced biomass, in one embodiment of the invention, the PHA-reduced biomass is returned back to the microorganism culture to be regenerated as PHA by a microorganism culture capable of utilizing PHA-reduced biomass, or a derivative thereof, as a source of carbon for PHA production, thereby creating a regenerative closed-loop polymerization system. By using PHA-reduced biomass as a source of carbon for PHA production in microorganisms growing as or in association with gas-utilizing microorganisms, PHA can be produced from carbon-containing gases at surprisingly and unexpectedly improved carbon, energy, and chemical efficiencies, since carbon from carbon-containing gases that would otherwise be discarded is regenerated as PHA in a microorganism culture, and microorganisms that produce PHA from carbon-containing gases at low concentrations (e.g., 5-60% PHA by weight, or less than 70% PHA by weight) can, in some embodiments, be utilized to produce PHA at signifi-

cantly increased carbon-to-PHA efficiencies. In some embodiments, the regeneration step is repeated to form an essentially closed-loop system. Thus, in some embodiments, the carbon output from the system is at least 1%, 5%, 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95%, or 99% PHA. In other words, at least 1%, 5%, 10%, 20%, 30%, 40%, 50%, 70%, 80%, 90%, 95%, or 99% of the carbon entering the system is converted into PHA. In other embodiments, 1-5%, 5-10%, 10-20%, 20-30%, 30%-40%, 40%-50%, 50%-60%, 60%-70%, 70%-80%, 80%-90%, 90%-95%, 95%-99% (and overlapping ranges thereof) of the carbon entering the system is converted to PHA. By regenerating PHA-reduced biomass as PHA in a microorganism culture, the percentage of carbon from a carbon containing gas that is converted to PHA is at least 2-fold, 3-fold, 4-fold, 5-fold, 6-fold, 7-fold, 8-fold, 9-fold, 10-fold, or greater than systems that do not employ the regenerative or closed-loop system disclosed herein. In some embodiments, the regeneration (e.g. return and/or recycling of the PHA-reduced biomass) step is repeated at least 2, 3, 4, 5, 6, 7, 8, 9, or 10) (or more) times. In some embodiments, the regeneration step is repeated until at least 90% to 95% of the carbon input into the system is converted into PHA. In some embodiments, the regeneration step is repeated as many times as desired to reach a particular percentage conversion of carbon to PHA.

[0016] Several embodiments of the invention provide for the production of PHA from carbon-containing gases at previously unattainable energy, carbon, and chemical efficiencies by way of providing a microorganism culture capable of metabolizing the carbon within both a carbon-containing gas and PHA-reduced biomass, manipulating the conditions of the culture to cause the culture to produce PHA, removing a portion of PHA-containing biomass from the culture, extracting the PHA within the removed PHA-containing biomass to create substantially isolated PHA and substantially PHA-reduced biomass, returning the PHA-reduced biomass to the culture and contacting the PHA-reduced biomass with the culture to cause the culture to metabolize the carbon within the PHA-reduced biomass into PHA, and purifying the isolated PHA. Thus, an advantage of several embodiments of the invention is the production of PHA from carbon-containing gases at significantly improved energy, carbon, and chemical efficiencies.

[0017] The process according to several embodiments disclosed herein yields a range of surprising benefits over current gas-based PHA production technologies. To begin, whereas the cell density of gas-based fermentation processes is traditionally limited by the mass transfer or diffusion rates of one or more factors, such as light, oxygen, carbon dioxide, methane, or volatile organic compounds, several embodiments disclosed herein enable the generation of cell densities that significantly exceed cell densities attainable in the current practice (e.g., by more than 1%, 10%, 20%, 30%, 50%, 80%, 100% or more), and thereby enables cost-efficient system mixing, aeration, heat control, and dewatering. For example, current methane-based PHA production systems are known to be capable (based on cell morphology and mass transfer characteristics) of generating approximately 60 g/L of biomass with an overall PHA concentration of 55%, or 33 g/L. PHA. In contrast, in several embodiments of the invention, cell densities of approximately 135 g/L with an overall PHA concentration of 70%, or 94.5 g/L. PHA are generated in a methane-based PHA production system. In some embodiments, cell densities of approxi-

mately 10 g/L, 20 g/L, 30 g/L, 60 g/L, 75 g/L, 100 g/L, 125 g/L, 135 g/L, 150 g/L or greater are achieved. In some embodiments, overall PHA concentration in such cultures ranges from approximately 1% to 20%, 20% to 30%, 30% to 55%, 55% to 60%, 65% to 70%, 70% to 80%, and overlapping ranges thereof result. In several embodiments, such PHA concentration ranges represent significant, unexpected, and surprising improvements over traditional processes, e.g., processes that are limited to low cell densities and/or PHA concentrations.

[0018] As a non-limiting example of the impact of this improvement on energy efficiency, the energy required, on an energy input-to-PHA output basis, to aerate, mix, and dewater a 135 g/L solution with a PHA concentration of 70% by weight is 186% less than the energy required to aerate, mix, and dewater a 60 g/L microorganism solution comprising 40% PHA by weight. It shall be appreciated that variations in the energy efficiency gains based on the systems and processes disclosed herein may occur, depending on the culture conditions, the strain or organisms used, and the initial gas stream or other carbon source. In several embodiments, even modest increases in efficiency have substantial benefits. For example, the ability to efficiently use an input gas having a low carbon concentration that would not otherwise be useful in PHA production may prevent the release of such a gas into the environment and/or reaction of the gas with other atmospheric compounds, thereby reducing the adverse impact of the low carbon concentration gas on the environment (e.g., destruction of ozone, greenhouse gas emission, pollution, etc.).

[0019] Additionally, whereas current gas-based PHA production systems produce significant carbon losses as a result of the low PHA inclusion concentrations of gas-utilizing microorganisms (i.e., a significant portion of carbon and energy input is lost as biomass), several embodiments of the invention enable the generation of overall carbon input yield efficiencies approaching maximum substrate values; e.g., 100% carbon input-to-PHA yield, minus respiration and/or downstream processing losses. In some embodiments, at least 5%, at least 10%, at least 30%, at least 50%, at least 70%, or at least 90%, carbon input-to-PHA yield is achieved. It is one important advantage of several embodiments of the invention that maximum carbon yield efficiencies are unexpectedly and surprisingly generated in a PHA production system employing gas-utilizing microorganisms, and particularly, in some embodiments, in PHA production systems employing gas-utilizing microorganisms that produce low biomass and/or PHA inclusion densities.

[0020] In some embodiments, the microorganism culture is a mixed culture, comprising heterotrophic microorganisms, methanotrophic microorganisms, autotrophic microorganisms, bacteria, yeast, fungi, algae, or combinations thereof. In other embodiments, the microorganism culture may be one or more cultures (e.g., a plurality of cultures). In some embodiments, the cultures are grown in one or more bioreactors. In some embodiments, the bioreactors utilize one or more culture conditions, including both aerobic and anaerobic conditions. In some embodiments, the microorganism culture converts PHA-reduced biomass to methane in an anaerobic process and subsequently to PHA in an aerobic process, such that PHA-reduced biomass is first anaerobically metabolized to methane and then used as methane to produce biomass and PHA in a methanotrophic culture.

[0021] In several embodiments, at least part of the microorganism culture is a mixed culture capable of metabolizing carbon-containing gases, including methane, carbon dioxide, greenhouse gases, and/or various other volatile organic compounds, into biomass and/or PHA. In some embodiments, the microorganism culture comprises a two phase system of anaerobic and anaerobic metabolism, whereby carbon-containing gas is produced in a first substantially anaerobic phase and subsequently converted into PHA in a second phase, wherein the microorganism culture in the first phase is substantially anaerobic and the culture in the second phase is either anaerobic or aerobic, wherein the two phases may be operated in one single vessel or in multiple vessels.

[0022] In some embodiments, at least one or more of the microorganisms are contacted with artificial and/or natural light during one or more steps of the methods disclosed herein.

[0023] In some embodiments, at least one of more of the microorganisms is contacted with dissolved oxygen during one or more steps of the methods disclosed herein.

[0024] In some embodiments, at least one of more of the microorganisms is cultured at atmospheric, sub-atmospheric, or above-atmospheric pressures.

[0025] In some embodiments, at least one of more of the microorganisms can utilize only a carbon-containing gas as a source of carbon.

[0026] In several embodiments, at least one of more of the microorganisms can utilize carbon derived from a PHA-reduced biomass as a source of carbon. In other embodiments, at least one or more of the microorganisms is a heterotrophic microorganism capable of converting PHA-reduced biomass into, carbon dioxide, oxygen, biomass, and/or PHA.

[0027] In several embodiments, at least one or more of the microorganisms are cultured using carbon derived from both a carbon-containing gas and a PHA-reduced biomass.

[0028] In some embodiments, the microorganism culture is a pure culture. In some embodiments, the cultures are maintained in semi-sterile or sterile conditions.

[0029] In some embodiments, the microorganism culture is a mixed, non-sterile culture, including a naturally equilibrating consortium of microorganisms.

[0030] In several embodiments, the microorganism culture is at least partially comprised of genetically engineered microorganisms.

[0031] In some embodiments, the microorganism culture is a mixed culture comprising a combination of naturally occurring and genetically engineered microorganisms.

[0032] In several embodiments, the PHA is removed from the microorganism culture by solvent extraction, including solvent extraction at temperatures ranging from 0° C. to 200° C. and at pressures ranging from -30 psi to 200 psi.

[0033] In several embodiments, the PHA is removed from the microorganism culture through the utilization of ketones, alcohols, and/or chlorinated solvents.

[0034] In several embodiments, the PHA is removed from the microorganism culture by hypochlorite digestion and/or chlorine-based solvent extraction.

[0035] In several embodiments, the PHA is removed from the microorganism culture by supercritical carbon dioxide extraction.

[0036] In several embodiments, the PHA is removed from the microorganism culture by protonic non-PHA cell material dissolution.

[0037] In several embodiments, the PHA is partially removed from the microorganism culture to create a PHA-rich phase and a PHA-poor phase.

[0038] In several embodiments, the PHA is removed from the microorganism culture to render the PHA substantially free of non-PHA material, including substantially 5%, 10%, 20%, 30%-40%, 40%-50%, 50%-60%, 60%-70%, 70%-80%, 80-90%, 90-99% or more pure PHA by weight.

[0039] In several embodiments, the PHA is removed from the microorganism culture by manipulating the pH of the microorganism culture.

[0040] In certain embodiments, at least one of more of the microorganisms are contacted with methane, carbon dioxide, oxygen, and/or a combination thereof.

[0041] In several embodiments, multiple culture vessels are employed, such that microorganism growth, PHA synthesis, PHA-reduced biomass metabolism, and PHA removal are carried out in separate vessels.

[0042] In other embodiments, microorganism growth, PHA-reduced biomass metabolism, and PHA synthesis occurs in a single vessel.

[0043] In still other embodiments, microorganism growth and PHA synthesis occur in a single vessel and PHA extraction is carried out in one or more separate vessels.

[0044] In several embodiments of the process as disclosed herein, PHA synthesis is regulated by manipulating the concentration of a material in the process, wherein the material is oxygen, methane, carbon dioxide, nitrogen, phosphorus, copper, iron, manganese, carbon, magnesium, potassium, cobalt, aluminum, sulfate, chlorine, boron, citric acid, or EDTA.

[0045] In several embodiments, the microorganism culture comprises one or more strains of microorganisms collectively capable of converting the carbon within a carbon-containing gas into cellular biomass and the carbon from cellular biomass or methane into PHA.

[0046] In several embodiments, the microorganisms are subjected to filtration, centrifugation, settling, and/or density separation.

[0047] In several embodiments, the isolated PHA and/or the PHA-reduced biomass is subjected to filtration, centrifugation, settling, and/or density separation.

[0048] In some embodiments, the process further comprises washing the recovered PHA with water, solvent, or other liquid-based agents to purify the PHA.

[0049] In several embodiments, the process further comprises oxidizing the recovered PHA to purify the PHA.

[0050] In several embodiments, the process further comprises drying the recovered PHA to remove volatiles such as water and/or one or more solvents.

[0051] In several embodiments of the invention, methods for the production of PHA are provided. In one embodiment, the method comprises. (a) providing a microorganism culture comprising PHA-containing biomass, (b) removing a portion of the PHA-containing biomass from the culture, (c) extracting a portion of the PHA from the removed PHA-containing biomass to produce isolated PHA and PHA-reduced biomass, (d) returning the PHA-reduced biomass to the culture to cause the culture to convert the carbon within the PHA-reduced biomass into PHA, and (e) purifying the isolated PHA.

[0052] In one embodiment, the microorganism culture utilizes the PHA-reduced biomass, or derivatives thereof, such as carbon dioxide, methane, or volatile organic acids,

volatile fatty acids, volatile organic compounds, non-methane organic compounds, and one or more carbon-containing gas as a source of carbon. In one embodiment, the gas is selected from the group consisting of methane, carbon dioxide, volatile organic compounds, and hydrocarbons. In one embodiment, the gas is derived from one or more sources from the group consisting of: landfills, wastewater treatment plants, power production facilities or equipment, agricultural digesters, oil refineries, natural gas refineries, cement production facilities, and/or anaerobic organic waste digesters

[0053] In some embodiments, the carbon in the PHA-reduced biomass is derived from one or more gases from the group consisting of: methane, biogas, carbon dioxide, volatile organic compounds, natural gas, wastewater treatment methane and VOCs, and hydrocarbons.

[0054] In some embodiments, natural and/or artificial light is utilized to induce the metabolism of the carbon dioxide by the culture.

[0055] In some embodiments, the microorganism culture comprises one strain, or a consortium of strains, of microorganisms, including one or more microorganisms selected from the group consisting of: bacteria, fungi, yeast, and algae, and combinations thereof.

[0056] In some embodiments, the microorganism culture comprises one or more microorganisms from the group consisting of: methanotrophic microorganisms, carbon-dioxide utilizing microorganisms, anaerobic microorganisms, methanogenic microorganisms, acidogenic microorganisms, acetogenic microorganisms, heterotrophic microorganisms, autotrophic microorganisms, cyanobacteria, and biomass-utilizing microorganisms, and combinations thereof.

[0057] In some embodiments, at least a portion of the microorganism culture is naturally occurring. In some embodiments, at least a portion of the microorganism culture is and/or genetically engineered. In some embodiments, naturally occurring and genetically engineered microorganisms are both used in the culture.

[0058] In some embodiments, the microorganism culture is at least partially maintained under above-atmospheric pressure.

[0059] In some embodiments, the PHA-containing biomass includes one or more microorganism-derived materials selected from the group consisting of: intracellular, cellular, and/or extracellular material, including a polymer, amino acid, nucleic acid, carbohydrate, lipid, sugar, polyhydroxyalkanoate, chemical, and/or metabolic derivative, intermediary, and/or end-product. In some embodiments, the PHA-containing biomass includes one or more microorganism-derived materials selected from the group consisting of methane, volatile organic compounds, carbon dioxide, and organic acids.

[0060] In one embodiment, the PHA-containing biomass contains less than about 95% water, including less than about 90%, 85%, 80%, 75%, or 70% water.

[0061] In some embodiments, the PHA-containing biomass is mixed with a chemical, including one or more chemicals from the group consisting of: methylene chloride, acetone, ethanol, methanol, ketones, alcohols, chloroform, and dichloroethane, or combinations thereof.

[0062] In one embodiment, the PHA-containing biomass is processed through homogenization, heat treatment, pH

treatment, enzyme treatment, solvent treatment, spray drying, freeze drying, sonication, and microwave treatment, or combinations thereof.

[0063] In one embodiment, the PHA-reduced biomass includes the PHA-containing biomass wherein at least a portion of the PHA has been removed from the PHA-containing biomass. In another embodiment, the PHA-reduced biomass includes methane, carbon dioxide, and organic compounds produced from the PHA-reduced biomass.

[0064] In some embodiments, the PHA-reduced biomass is subject to dewatering, chemical treatment, sonication, additional PHA extraction, homogenization, sonication, heat treatment, pH treatment, hypochlorite treatment, microwave treatment, microbiological treatment, including both aerobic and anaerobic digestion, solvent treatment, water washing, solvent washing, and/or drying, including simple or fractional distillation, spray drying, freeze drying, and/or oven drying, or combinations thereof.

[0065] In several embodiments, the microorganism culture is maintained in a sterile, semi-sterile, or non-sterile environment.

[0066] In one embodiment, the PHA includes one or more PHA selected from the group consisting of: polyhydroxybutyrate (PHB), polyhydroxyvalerate (PHV), polyhydroxybutyrate-covalerate (PHB/V), polyhydroxyhexanoate (PHHx), and short chain length (SCL), medium chain length (MCL), and long chain length (LCL) PHAs.

[0067] In several embodiments, the metabolism, growth, reproduction, and/or PHA synthesis of the culture is controlled, manipulated, and/or affected by a growth medium. In some embodiments, the bioavailable and/or total concentration of nutrients within the growth medium, such as copper, iron, oxygen, methane, carbon dioxide, nitrogen, magnesium, potassium, calcium, phosphorus, EDTA, calcium, sodium, boron, zinc, aluminum, nickel, sulfur, manganese, chlorine, chromium, molybdenum, and/or combinations thereof are manipulated (e.g., increased, decreased, or maintained) in order to control the metabolism, growth, reproduction, and/or PHA synthesis of the culture. In some embodiments, a single nutrient in the growth medium is manipulated, while in some embodiments, more than one nutrient in the growth medium is manipulated to achieve the desired effect on the culture.

[0068] In one embodiment, the conversion of the PHA-reduced biomass into the PHA is induced and/or controlled by manipulating the composition of the medium. As discussed herein, the conversion of PHA-reduced biomass into the PHA can be controlled in a time-dependent manner to maximize the efficiency of conversion. In some embodiments, conversion to PHA production is induced about 1-12 hours, about 5-15 hours, or about 8-24 hours after PHA-reduced biomass is re-introduced into the culture. In some embodiments, longer times, e.g., about 24 hours to several days or weeks, are employed.

[0069] In one embodiment, the conversion of the PHA-reduced biomass into the PHA is effected by manipulating the concentration one or more elements selected from the group consisting of: nitrogen, methane, carbon dioxide, phosphorus, oxygen, magnesium, potassium, iron, copper, sulfate, manganese, calcium, chlorine, boron, zinc, aluminum, nickel, and/or sodium, and combinations thereof.

[0070] In some embodiments, the PHA is at least partially removed from the PHA-containing biomass using one or

more extraction agents selected from the group consisting of: solvents, including methylene chloride, acetone, ethanol, methanol, or dichloroethane, supercritical carbon dioxide, sonication, homogenization, water, heat, distillation, spray drying, freeze drying, enzymes, surfactants, acids, bases, hypochlorite, peroxides, bleaches, ozone, EDTA, and/or combinations thereof.

[0071] In one embodiment, the extraction process is substantially carried out at intracellular temperatures less than 100° C. In other embodiments, temperatures for extraction range from about 10° C. to 30° C., from about 30° C. to 50° C., from about 50° C. to 70° C., from about 70° C. to 90° C., from about 90° C. to about 120° C., or higher. In another embodiment, cells are reused for polymerization following the extraction process as viable cells.

[0072] In one embodiment, the removal of the PHA from the culture causes the culture to be temporarily deactivated, such that the culture, or elements thereof, may be further used for the synthesis of PHA. In certain embodiments, deactivation is beneficial because it allows for the delay of PHA production, transfer of material to another production area, and the like. In some embodiments, deactivation allows a tailored PHA production time frame. In some embodiments, the reuse of cells for polymerization is beneficial because it avoids or reduces the need to produce new biomass prior to polymerization, thereby reducing the carbon, chemical, and energy requirement of PHA production.

[0073] In one embodiment, a PHA produced according to the several embodiments described herein is provided.

[0074] In several embodiments, processes for the production of PHA from a carbon-containing gas are provided. In one embodiment, the process comprises the steps of: a) providing a growth medium comprising a microorganism culture capable of utilizing the carbon within one or more carbon-containing gas and PHA-reduced biomass, b) manipulating the medium to cause the culture to produce PHA, c) removing at least a portion of the PHA within the culture to create substantially isolated PHA and substantially PHA-reduced biomass, d) purifying the isolated PHA, and e) returning the PHA-reduced biomass to the culture to cause the culture to metabolize the carbon within the PHA-reduced biomass into PHA.

[0075] In one embodiment, the carbon-containing gas is selected from the group consisting of: methane, carbon dioxide, toluene, xylene, butane, ethane, methylene chloride, acetone, ethanol, propane, methanol, vinyl chloride, volatile organic compounds, hydrocarbons, and combinations thereof.

[0076] In one embodiment, the invention comprises a PHA comprising carbon derived from PHA-reduced biomass, wherein the PHA-reduced biomass comprises carbon derived from one or more carbon-containing gas.

[0077] In one embodiment, a PHA derived from a carbon-containing greenhouse gas, including methane, carbon dioxide, or combinations thereof, is provided. In some embodiments, use of such a gas is particularly advantageous, as it allows for the simultaneous production of PHA at lower energy costs and higher efficiencies, but also removes a portion of a destructive gas from the atmosphere. In some embodiments, processes and systems as disclosed herein are particularly well suited for use near sources of such gases (e.g., landfills, power production plants, anaerobic digesters, etc.) for onsite conversion of harmful gasses to a commercially valuable product.

[0078] In several embodiments, processes for the oxidation of methane are provided. In one embodiment, the process comprises: providing a culture of methanotrophic and autotrophic microorganisms, providing a growth culture medium comprising dissolved methane and carbon dioxide, and contacting the culture with light to cause the culture to convert the carbon dioxide into oxygen, whereby the culture utilizes the oxygen to oxidize the methane, thereby reducing or eliminating the need for an extraneous source of oxygen to drive methanotrophic metabolism.

[0079] In some embodiments, the light used to contact the culture is artificial light. In some embodiments, the light used to contact the culture is natural light. In other embodiments, combinations of natural and artificial light are used. In some such embodiments, wavelengths of artificial light are specifically filtered out or controlled such that the culture is exposed to a broader or more controlled overall spectrum of light (e.g., the sum of wavelengths of natural light and artificial light). In some embodiments, the source of light also functions to generate heat, which can be used to maintain optimal culture temperatures. In other embodiments, light input is regulated by time, such that specific cultures of autotrophic and/or heterotrophic microorganisms are selected for or optimized according to the duration and/or pattern of light injection (e.g., 0-12 or 12-24 hours light injection, 0-12 or 12-24 hours dark incubation, multi-second pulsation, etc.).

[0080] In one embodiment, the addition of light reduces the need for exogenous oxygen sources. While such embodiments provide an advantage in reducing costs of input materials, in some embodiments, an exogenous source of oxygen, including air, is added to the culture.

[0081] In some embodiments the addition of autotrophic microorganisms to the culture impacts the metabolism of the culture. In such an embodiment, the timed and planned addition, activation, or metabolic enhancement of autotrophic organisms can be based on the desire for changing the rate of methane oxidation.

[0082] In some embodiments, a system is used for PHA production comprising providing i) a culture of autotrophic, methanotrophic, methanogenic, and/or heterotrophic microorganisms and ii) a first gas comprising carbon dioxide, methane, volatile organic compounds, oxygen, and/or other gas, whereby the culture of microorganisms are caused to use the first gas to generate a second gas comprising carbon dioxide, methane, oxygen, volatile organic compounds, and/or other gas, whereby the culture subsequently is caused to utilize the second gas for the generation of PHA, which can then be isolated and purified according to several embodiments disclosed herein. In some embodiments, the first gas can be methane, carbon dioxide, oxygen, or volatile organic compounds. In other embodiments, the second gas can be oxygen, methane, carbon dioxide, or other volatile organic compounds. In some embodiments, microorganisms can be used to convert carbon dioxide to biomass which can in turn be used to produce methane, which can be subsequently used to produce PHA. In other embodiments, microorganisms can be used to convert carbon dioxide to oxygen which can in turn be used to produce PHA. As disclosed herein, the products generated at each of these steps may be recycled (e.g., splitting a portion of the autotrophic culture and recycling it to generate additional biomass, generating

reduced-PHA biomass and recycling it into the methanotrophic culture to generate additional biomass and additional PHA).

[0083] In several embodiments, processes for producing autotrophic microorganisms using only methane as a carbon input are provided. In one embodiment, the process comprises: adding methane, oxygen, and methane-utilizing microorganisms to a culture of autotrophic microorganisms, whereby the methane-utilizing microorganisms convert the methane into carbon dioxide, and whereby the autotrophic microorganisms utilize the carbon dioxide as a source of carbon, thereby reducing or eliminating the need for an extraneous source of carbon dioxide to drive autotrophic metabolism. In some embodiments, addition of methanotrophic and/or heterotrophic microorganisms to the culture impacts the metabolism of the autotrophic microorganisms. As discussed herein, the purposeful addition of such microorganisms at particular times allows for specific levels of control over the overall output and operation of the system.

[0084] In several embodiments, processes for oxidizing methane at low concentrations are provided. In one embodiment, the process comprises: culturing methanotrophic microorganisms in a medium comprising water, dissolved methane, dissolved oxygen, and mineral salts, adding methanol to the medium at a rate and volume sufficient to cause the microorganisms to reduce the concentration of the methane in the medium, whereby substantially all of the methane within the medium is utilized, thereby enabling methanotrophic microorganisms to metabolize methane present at low bioavailable concentrations. In some embodiments, gas containing less than 20% methane by volume is contacted with the medium. In some embodiments, gas containing less than 1% methane by volume is contacted with the medium. In some embodiments, the methanol is produced by microorganism metabolism.

[0085] In several embodiments, processes for separating water from microorganism biomass are provided. In one embodiment, the process comprises: providing biomass mixed with water in a liquid medium, mixing the medium with a liquid agent selected from the group consisting of ketones, alcohols, chlorinated solvents, derivatives thereof, or combinations thereof, and subjecting the mixture to a filtration step. In several embodiments, this enables the efficient separation of biomass from water. In some embodiments, such methods reduce or eliminate the need for centrifugation in the separation process. In some embodiments, the liquid agent is acetone, ethanol, isopropanol, and/or methanol. In other embodiments, other liquid agents that are miscible with water are used to separate the biomass from the aqueous portion of the mixture. In several embodiments separation is achieved by centrifugation (high-speed, low-speed), gravity separation, multi-stage filtration, or combinations thereof.

[0086] In several embodiments, processes for extracting a polyhydroxyalkanoate from a PHA-containing biomass are provided. In one embodiment, the process comprises the steps of: (a) providing a PHA-containing biomass comprising PHA and water, (b) mixing said biomass with a solvent at a temperature sufficient to dissolve at least a portion of said PHA into said solvent and at a pressure sufficient to enable substantially all or part of said solvent to remain in liquid phase, thereby producing a PHA-lean biomass phase and a PHA-rich solvent phase comprising water, PHA and solvent (c) separating said PHA-rich solvent phase from said

PHA-lean biomass phase at a temperature and pressure sufficient to enable substantially all or part of said solvent to remain in liquid phase and prevent substantially all or part of said PHA within said PHA-rich solvent phase from precipitating into said water, (d) reducing the pressure or increasing the temperature of said PHA-rich solvent phase to cause said PHA-rich solvent to vaporize and said PHA to precipitate or otherwise become a solid PHA material while maintaining the temperature and/or pressure of the PHA-rich solvent phase to prevent all or part of the temperature-dependent precipitation of said PHA into said water, and (e) collecting said solid PHA material, including optionally separating said solid PHA material from said solvent and/or said water.

[0087] In some embodiments, suitable solvents include acetone, ethanol, methanol, dichloroethane, and/or methylene chloride. Depending on the solvent selected, in some embodiments, separating the solid PHA material from solvent and/or water is achieved by increasing the temperature of the mixture. In other embodiments, separation is achieved through reducing the pressure of the solvent, PHA, and/or water. In some embodiments, combinations of temperature changes and pressure changes are used to optimally separate solid PHA material from solvent and/or water. In some embodiments, evaporation of solvent and/or water occurs in a rapid fashion, thereby reducing the need for temperature or pressure changes. Advantageously, certain embodiments of the processes disclosed herein may optionally be carried out in a batch, semi-continuous, or continuous manner. Thus, the process can be tailored to the needs of the producer at any given time.

[0088] In several embodiments, processes for modifying the functional characteristics of a PHA are provided. In one embodiment, the process comprises providing a first PHA and a second PHA, wherein the molecular weight of said second PHA is greater than the molecular weight of said first PHA, and combining said first PHA with said second PHA to modify the functional characteristics of both said first PHA and second PHA. In some embodiments, both the first PHA and the second PHA are PHB, and in some embodiments, one or more of the first and second PHA comprises PHB/V. In one embodiment the first PHA and the second PHA is PHB or PHBV.

[0089] In some embodiments, the molecular weight of the first PHA is greater than about 500,000 Daltons and the molecular weight of the second PHA is less than about 500,000 Daltons. However, in some embodiments, the molecular weight can be adjusted. For example, in some embodiments, a first PHA is subjected to a temperature sufficient to reduce the molecular weight of the first PHA. Thereafter, it can be combined with the second PHA. It shall be appreciated that the second PHA could also optionally be exposed to temperature in order to adjust its molecular weight. In some embodiments, the molecular weight of the second PHA is greater than about 800,000 Daltons. In certain embodiments, the molecular weight of said second PHA is greater than about 1,000,000 Daltons. In some embodiments, the molecular weights of the first and second PHA are specifically tailored relative to one another, (e.g., a ratio of 1:2, 1:4, 1:6, 1:8, 1:10, etc.) in order to maximize the alterations in functional characteristics.

[0090] In several embodiments, processes for increasing the penetration depth of light in a liquid are provided. In one embodiment, the process comprises the steps of (a) directing

light into a liquid medium in the form of a light path and (b) reducing the density of liquid in the light path.

[0091] In several embodiments, the density of the liquid in the light path is reduced by adding gas to said liquid in the light path. In some embodiments, the gas is air, oxygen, methane, carbon dioxide, nitrogen, and/or a combination thereof. In some embodiments, the gas is simultaneously added along with the light. In certain embodiments, the gas and the light are emitted or injected into the liquid through a common material, such as a permeable or semi-permeable membrane through which light and/or gas traverse. In other embodiments, the light and the gas are added separately. In such embodiments, customization of the addition is possible. For example, gas can be added in pulses (e.g., on/off sequences), continuously, or in bracketed time frames around the addition of light. In some embodiments, the addition of light and gas are coordinated to maximize the penetration of the light. For example a burst of gas followed by a burst of light (or overlapping to some degree) may advantageously increase the penetration of the light.

[0092] In several embodiments, processes for modifying the pH in a microorganism culture medium are provided. In one embodiment, the process comprises the steps of: (a) providing a culture medium comprising water and microorganisms, (b) adding a first source of nitrogen to the medium to cause the microorganisms to metabolize the nitrogen and thereby increase the concentration of either hydroxyl ions or protons, respectively, in the medium, and (c) adding a second source of nitrogen to the medium to cause the microorganisms to metabolize the nitrogen and thereby increase the concentration of either protons or hydroxyl ions, respectively, in the medium. In other embodiments, a source of nitrogen is added to the culture that increases the pH of the medium, wherein the metabolism of the nitrogen source causes the pH of the medium to decrease, thereby reducing or eliminating the need for an additional pH adjustment step.

[0093] In several embodiments, low shear processes for adding gas to a microorganism culture medium are provided. In one embodiment, the process comprises the steps of: (a) providing a liquid medium and a gas, (b) contacting the medium with the gas in a first container to cause at least a portion of the gas to dissolve in the medium, (c) providing a second container, and (d) transferring at least a portion of the liquid comprising the gas within the first container to the second container. In some embodiments, a mixer is also provided, in order to dissolve a portion of the gas in the medium. In some embodiments, the mixer is a pump or agitator or high shear mixer. In some embodiments, the mixer comprises a centrifugal pump. In still other embodiments, the gas itself provides a mixing function. For example, the injection of gas into a medium will result in gas bubbles, which, if released at the bottom of a container comprising medium, will not only promote the dissolution of gas into the medium, but mix the medium as the bubbles rise.

[0094] In several embodiments, processes for injecting gas into a pressurized microorganism culture vessel are provided. In one embodiment, the process comprises the steps of providing a vessel comprising a medium comprising microorganisms, adding a gas into the vessel that can be metabolized by the microorganisms, and adjusting the flow rate of the gas into the vessel according to the rate of change of pressure within the vessel. In some embodiments, the gas

is oxygen, methane, carbon dioxide, or combinations thereof. Choice of the gas depends on the vessel used and the culture within the vessel. In some embodiments, backpressure monitoring allows for optimal gas injection for a given culture (e.g., if certain cultures react more quickly to administration of a gas and rapidly increase pressure, flow can be coordinately reduced).

[0095] In one embodiment, a process for producing light in a liquid medium is provided. In one embodiment, the process comprises the steps of: a) providing a liquid, b) providing a light-emitting unit or material comprising two conductive leads and a light-emitting juncture between the conductive leads, or a material that will emit light when contacted with electrons and c) inducing a voltage in the liquid, thereby inducing the movement of electrons in the conductive leads of the light-emitting unit or inducing electrons to contact the material, thereby causing the light-emitting unit or material to emit light. In some embodiments, the material is a phosphor, phosphoric, and/or luminescent material, including an electroluminescent phosphor.

[0096] In one embodiment, AC voltage is induced in a liquid by inserting the two leads of a 115V AC power source into a liquid. Without being bound by theory, it is believed that a liquid carrying an AC voltage is capable of inducing the movement of electrons into and through a light-emitting device suspended in the liquid and not contacting the two 115V AC power source leads due to the oscillating nature of electrons in an AC circuit, such that AC voltage in a liquid causes electrons to fill conductive paths connected to the liquid, in spite of the resistance of the conductive paths relative to the liquid, and will oscillate as an AC current in those conductive paths, thereby performing work, e.g., generating light in a light emitting diode. As a non-limiting example, light is produced in a liquid medium by a) placing a light-emitting diode in a liquid comprising water and electrolytic ions, and b) inducing an AC voltage in the liquid, wherein c) the induction of AC voltage in the electrolytic liquid causes the light-emitting diode to generate light.

[0097] Through experimentation, Applicant unexpectedly discovered that one or multiple light emitting units will emit light in a liquid when AC voltage is applied to the liquid and when the light-emitting units are rated for voltages and amp draws commensurate with available electrical energy. The production of autotrophic microorganisms is fundamentally constrained by the ability of light to penetrate through a liquid and thereby enable photosynthesis. Prior to Applicant's invention as disclosed herein, no methods were believed to be known to produce light in a liquid through the utilization of light-emitting devices physically unconnected to an electrical voltage source vis-à-vis solid conductive material. In one embodiment, the utilization of one or more, and preferably many, free-floating light-emitting units in an electrically charged liquid comprising autotrophic microorganisms enables a very high light transmission efficiency, wherein previous light penetration constraints are largely overcome and high autotrophic microorganism densities are fully enabled.

[0098] In several embodiments, a method for producing a polyhydroxyalkanoate (PHA) in a microorganism culture is provided. In some embodiments, the method comprises the steps of: a) subjecting said culture to a growth period comprising exposing said culture to growth conditions to cause said culture to reproduce, (b) subjecting said culture to a polymerization period comprising exposing said culture to

polymerization conditions to cause said culture to produce intracellular PHA, and (c) repeating step (a) and then second step (b) two or more times.

[0099] In some embodiments the growth period comprises a period in which the culture reproduces or otherwise produces biomass and/or reproduces. In some embodiments, the polymerization period comprises a period in which the culture synthesizes PHA. In some embodiments, the growth period and the polymerization period are induced by the culture media (e.g., the extracellular media around the culture). In some embodiments, alteration in the media conditions induce a transition (partial or complete) between growth and polymerization periods. In some embodiments, a culture is cycled between growth and polymerization periods two, three, four, or more times, in order to produce PHA and then reproduce biomass, which is subsequently used to generate additional PHA.

[0100] In some embodiments, the culture is exposed to non-sterile conditions. In certain such embodiments, input carbon is non-sterile. However, in some embodiments, sterile conditions exist. In some embodiments, the culture is dynamic over time in that it may be exposed to extraneous microorganisms. In certain embodiments, this is due to a non-sterile culture environment. In some embodiments, extraneous microorganisms potentiate the production of PHA.

[0101] In several embodiments, a process for the reduction of pigmentation in a microorganism culture is provided. In one embodiment, the process comprises the steps of: providing a medium comprising a microorganism culture comprising dissolved oxygen, and increasing the concentration of dissolved oxygen over successive periods to select for light-colored or low-level pigmentation microorganisms.

[0102] PHA, while able to be treated post-production to reduce pigmentation, is less expensive to produce when lower levels of pigmentation exist. Some microorganisms are more pigmented than others, and therefore in several embodiments, selection against the more pigmented microorganisms results in a less pigmented PHA, which reduces production costs. In some embodiments, the microorganisms cultured and selected for are methanotrophic microorganisms. By manipulating the culture conditions, which benefit certain varieties of microorganisms, a less pigmented culture (and hence a less pigmented PHA) result. In some embodiments, increases in concentration of dissolved oxygen over periods ranging from 1-3 hours, 3-5 hours, 5-7 hours, 7-10 hours, 10-15 hours, or 15-24 hours are used to select for less pigmented microorganisms.

[0103] In several embodiments, a process for the conversion of a gas into a polyhydroxyalkanoate (PHA) is provided, wherein the process comprises the steps of: a) providing i) a first gas and ii) a culture of microorganisms, b) contacting the first gas with the culture to cause the culture to convert the first gas into a second gas c) contacting the second gas with the culture, and d) causing the culture to use the second gas to produce PHA.

[0104] In some embodiments, the microorganisms are autotrophic, methanogenic, heterotrophic, or combinations thereof.

[0105] In one embodiment the first gas is carbon dioxide. In one embodiment the first gas is oxygen. In one embodiment the first gas is methane. In one embodiment the first gas is a volatile organic compound.

[0106] In one embodiment the second gas is carbon dioxide. In one embodiment the second gas is oxygen. In one embodiment the second gas is methane. In one embodiment the second gas is a volatile organic compound.

[0107] It shall be appreciated that the selection of the first and the second gas is based on the type of microorganism or microorganisms being cultured.

BRIEF DESCRIPTION OF THE DRAWING

[0108] FIG. 1 is a block flow diagram comprising the steps of: microorganism fermentation and PHA synthesis, PHA-containing biomass removal, PHA-reduced biomass and isolated PHA production, PHA-reduced biomass recycling and fermentation, and isolated PHA purification.

DETAILED DESCRIPTION

[0109] While PHAs have significant environmental advantages compared to fossil fuel-based plastics, the cost of PHA production is generally viewed as a significant limitation to the industrial production and commercial adoption of PHAs. Generally, the overall cost of PHA production is determined by three major inputs: 1) carbon, 2) chemicals, and 3) energy. Accordingly, efforts to reduce the cost of PHA production must address one or more of these areas, specifically by: i) reducing carbon input costs, ii) increasing carbon-to-PHA yields, iii) reducing the volume of chemicals required for PHA production, and/or iv) increasing energy-to-PHA yields.

[0110] As discussed above, food crop derived sugars in genetically engineered microorganism-based aqueous fermentation systems are widely regarded as the most carbon, chemical, energy, and, thus, cost efficient PHA production method. Despite these efficiencies, sugar-based PHA production remains many times more expensive than fossil fuel-based plastics production. Attempts to reduce the carbon input cost of the PHA production process, by utilizing carbon-containing industrial off-gases, such as carbon dioxide and methane, have been previously limited by technical challenges and stoichiometric limitations that render the gas-to-PHA production process significantly more energy and chemical intensive, and thus more costly, than the food crop-based PHA production process.

[0111] Specifically, these technical challenges and stoichiometric limitations include: low mass transfer rates, low microorganism growth rates, extended polymerization times, low cell densities, high oxygen demand (relative to solid substrates), and low PHA cellular inclusion concentrations. Whereas sugar-based fermentation systems have the ability to generate high cellular densities and PHA inclusion concentrations, carbon-containing gas-based fermentation processes typically cannot, based on fundamental cell morphology and mass transfer constraints, generate cellular and PHA densities exceeding 10-30% of densities possible in sugar-based processes. As a result, the ratio of energy-to-PHA required to carry out upstream carbon injection, oxygen injection, system cooling, and culture mixing, as well as downstream PHA purification, significantly exceeds the energy-to-PHA ratio required for sugar-based PHA production methods, thereby rendering the emissions-based process uncompetitive when compared to both petroleum-based plastics and sugar-based PHAs.

[0112] Several embodiments of the present invention therefore relate to a novel method for the production of PHA

using carbon-containing gases as a source of carbon, wherein the energy input-to-PHA production ratio, carbon input-to-PHA production ratio, and cost efficiency of the process is significantly improved over previous gas-based PHA production processes.

[0113] In several embodiments, this process may be accomplished by a) culturing a first microorganism culture capable of metabolizing the carbon within both a carbon-containing gas and biomass, or a derivative thereof, b) manipulating the conditions of the culture to cause the culture to produce PHA-containing biomass, c) removing a portion of the PHA-containing biomass; d) extracting at least a portion of the PHA within the removed PHA-containing biomass to create substantially isolated PHA and substantially PHA-reduced biomass, e) purifying the isolated PHA, and f) returning the PHA-reduced biomass to the microorganism culture to cause the microorganism culture to metabolize the carbon within the PHA-reduced biomass into PHA.

[0114] According to some embodiments, the steps of this process are as follows: (a) providing a microorganism culture comprising biomass and PHA; (b) removing a portion of the PHA-containing biomass from the culture, and extracting PHA from the removed PHA-containing biomass to produce isolated PHA and PHA-reduced biomass; (c) purifying the isolated PHA, and (d) returning the PHA-reduced biomass to be mixed with the culture to cause the culture to convert the carbon within the PHA-reduced biomass into PHA. Each of the above recited steps in the process are discussed in more detail below.

Providing a Microorganism Culture Comprising Biomass and PHA

[0115] The terms “microorganism”, “microorganisms”, “culture”, “cultures”, and “microorganism cultures”, as used herein, shall be given their ordinary meaning and shall include, but not be limited to, a single microorganism and/or consortium of microorganisms, including, among others, genetically-engineered bacteria, fungi, algae, and/or yeast. In some embodiments, microorganisms are naturally occurring and in some embodiments microorganisms are genetically-engineered. In some embodiments, both naturally occurring and genetically-engineered microorganisms are used. In some embodiments, a mixed culture of microorganisms may be used. In some embodiments, microorganisms or cultures shall include a microorganism metabolism system, including the interactions and/or multiple functions of multiple cultures in one or more conditions.

[0116] The terms “biomass” and “biomass material” shall be given their ordinary meaning and shall include, but not be limited to, microorganism-derived material, including intracellular, cellular, and/or extracellular material, such materials including, but not limited to, a polymer or polymers, amino acids, nucleic acids, carbohydrates, lipids, sugars, PHA, volatile fatty acids, chemicals, gases, such as carbon dioxide, methane, volatile organic acids, and oxygen, and/or metabolic derivatives, intermediaries, and/or end-products. In several embodiments, biomass is dried or substantially dried.

[0117] In some embodiments, the biomass contains less than about 99% water. In other embodiments, the biomass contains between about 99% to about 75% water, including about 95%, 90%, 85%, and 80%. In some embodiments, the biomass contains between about 75% and about 25% water,

including 75%-65%, 65%-55%, 55%-45%, 45%-35%, 35%-25%, and overlapping ranges thereof. In additional embodiments, the biomass contains from about 25% water to less than about 0.1% water, including 25%-20%, 20%-15%, 15%-10%, 10%-5%, 5%-1%, 1%-0.1%, and overlapping ranges thereof. In still other embodiments, the biomass contains no detectable amount of water. Depending on the embodiment, water is removed from the biomass by one or more of freeze drying, spray drying, fluid bed drying, ribbon drying, flocculation, pressing, filtration, and/or centrifugation. In some embodiments, the biomass may be mixed with one or more chemicals, such as methylene chloride, acetone, methanol, and/or ethanol, at various concentrations. In other embodiments, the biomass may be processed through homogenization, heat treatment, pH treatment, enzyme treatment, solvent treatment, spray drying, freeze drying, sonication, or microwave treatment. As used herein, the term “PHA-reduced biomass” shall be given its ordinary meaning and shall mean any biomass wherein at least a portion of PHA has been removed from the biomass through a PHA extraction process. As used herein, the term “PHA-containing biomass” shall be given its ordinary meaning and shall mean any biomass wherein at least a portion of the biomass is PHA.

[0118] Microorganism cultures useful for the invention described herein include a single strain, and/or a consortium of strains, which are individually and/or collectively capable of using carbon containing gases and biomass, including PHA-reduced biomass, as a source of carbon for the production of biomass and PHA. In some embodiments, a microorganism culture according to several embodiments, comprises a microorganism culture that utilizes PHA-reduced biomass, or any derivative thereof, including methanotrophic microorganisms, anaerobic digestion cultures, and other heterotrophic microorganisms, as a source of carbon for the production of biomass, or metabolic derivatives including, and in particular, the production of PHA, protein, methane, and/or carbon dioxide (herein, “biomass-utilizing microorganisms”). As used herein, the terms “microorganism”, “culture”, “microorganism culture,” “microorganism system,” “microorganism consortium,” and “consortium of microorganisms” are used interchangeably. Additionally, any of these terms may refer to one, two, three, or more microorganism cultures and/or strains, including a microorganism system that is collectively capable of carrying out a complex metabolic function (e.g., conversion of PHA-reduced biomass to methane, carbon dioxide, protein, and/or PHA). In several embodiments, the microorganism culture comprises of a consortium of carbon-containing gas-utilizing microorganisms and a consortium of biomass-utilizing microorganisms. In some embodiments, the gases metabolized by such cultures comprise methane, carbon dioxide, and/or a combination thereof.

[0119] In some embodiments, the microorganism culture comprises a consortium of acidogenic, acetogenic, methanogenic, methanotrophic, and/or autotrophic microorganisms in one or more individual bioreactors. As such, in some embodiments, the cultures are grown in one or more distinct culture conditions. In some embodiments, the conditions are either aerobic or anaerobic conditions. In some embodiments, culture conditions are varied over time (e.g. initially aerobic with a transition to anaerobic, or vice versa). As used herein, the term “bioreactor” shall be given its ordinary

meaning and shall also refer to a tank, vessel, or any container or device suitable for growth and culturing of microorganisms.

[0120] In some embodiments, the microorganism culture is contained within a single vessel, wherein the steps of converting PHA-reduced biomass to biomass, converting biomass to PHA, and converting carbon-containing gases to biomass and/or PHA occur simultaneously or sequentially.

[0121] In other embodiments, the microorganism culture is contained within multiple vessels, which are designed to carry out specific and unique functions. For example, one embodiment includes the steps of (a) converting PHA-reduced biomass to PHA-reduced biomass-derived materials such volatile organic acids, methane, and/or carbon dioxide, which is carried out in a first vessel and (b) synthesizing PHA from PHA-reduced biomass-derived materials which is carried out in a second, separate tank under independent conditions. In some embodiments, one or more of the tanks is an anaerobic digestion tank and one or more other tank is an aerobic fermentation tank.

[0122] As used herein, the term “gas-utilizing microorganisms” shall be given its ordinary meaning and shall refer to microorganisms capable of utilizing gases containing carbon for the production of biomass, including the production of PHA. Similarly, the terms “methanotrophic microorganisms” and “methane-utilizing microorganisms” shall be given their ordinary meanings and shall refer to microorganisms capable of utilizing methane as a source of carbon for the production of biomass. Further, the terms “Autotrophic microorganisms” and “carbon dioxide-utilizing microorganisms” shall be given their ordinary meaning and shall refer to microorganisms capable of utilizing carbon dioxide as a source of carbon for the production of biomass, including microorganisms that utilize natural and/or synthetic sources of light to carry out the metabolism of carbon dioxide into biomass. The term “heterotrophic microorganisms”, as used herein, shall be given its ordinary meaning and shall include methanotrophic, methanogenic, acidogenic, acetogenic and biomass-utilizing microorganisms, including microorganisms that convert sugar, volatile fatty acids, or other carbon substrates to biomass. The term “methanogenic microorganisms” shall be given its ordinary meaning and shall refer to microorganisms that convert biomass to methane, including the consortium of microorganisms required to carry out such a process, including, but not limited to, acidogenic and acetogenic microorganisms.

[0123] As discussed herein, in several embodiments, carbon-containing gases are used as a source of carbon by microorganism cultures. In some embodiments, other sources of carbon are used (e.g., PHA-reduced biomass), either alone or in combination with carbon-containing gases. In some embodiments, the carbon-containing gases used include, but are not limited to, carbon dioxide, methane, ethane, butane, propane, benzene, xylene, acetone, methylene chloride, chloroform, volatile organic compounds, hydrocarbons, and/or combinations thereof. The source of the carbon-containing gases depends on the embodiment. For example, carbon-containing gas sources used in some embodiments include landfills, wastewater treatment plants, anaerobic metabolism, power production facilities or equipment, agricultural digesters, oil refineries, natural gas refineries, cement production facilities, and/or anaerobic organic material digesters, including both solid and liquid material digesters.

[0124] In several embodiments described herein, microorganisms may include, but are not limited to, yeast, fungi, algae, and bacteria (including combinations thereof). Suitable yeasts include, but are not limited to, species from the genera *Candida*, *Hansenula*, *Torulopsis*, *Saccharomyces*, *Pichia*, *1-Debaryomyces*, *Lipomyces*, *Cryptococcus*, *Nematosporea*, and *Brettanomyces*. Suitable genera include *Candida*, *Hansenula*, *Torulopsis*, *Pichia*, and *Saccharomyces*. Examples of suitable species include, but are not limited to: *Candida boidinii*, *Candida mycoderma*, *Candida utilis*, *Candida stellatoidea*, *Candida robusta*, *Candida clausenii*, *Candida rugosa*, *Brettanomyces petrophilum*, *Hansenula minuta*, *Hansenula satumus*, *Hansenula californica*, *Hansenula mrakii*, *Hansenula silvicola*, *Hansenula polymorpha*, *Hansenula wickerhamii*, *Hansenula capsulata*, *Hansenula glucozyma*, *Hansenula henricii*, *Hansenula nonfermentans*, *Hansenula philodendra*, *Torulopsis candida*, *Torulopsis bolmii*, *Torulopsis versatilis*, *Torulopsis glabrata*, *Torulopsis molishiana*, *Torulopsis nemodendra*, *Torulopsis nitrato-phila*, *Torulopsis pinus*, *Pichia farinosa*, *Pichia polymorpha*, *Pichia membranaefaciens*, *Pichia pinus*, *Pichia pastoris*, *Pichia trehalophila*, *Saccharomyces cerevisiae*, *Saccharomyces fragilis*, *Saccharomyces rosei*, *Saccharomyces acidifaciens*, *Saccharomyces elegans*, *Saccharomyces rouxii*, *Saccharomyces lactis*, and/or *Saccharomyces fructum*.

[0125] Suitable bacteria include, but are not limited to, species from the genera *Bacillus*, *Mycobacterium*, *Actinomyces*, *Nocardia*, *Pseudomonas*, *Methanomonas*, *Methylo-monas*, *Brevibacterium*, *Protaminobacter*, *Methylococcus*, *Arthrobacter*, *Acetobacter*, *Methylomonas*, *Brevibacterium*, *Acetobacter*, *Micrococcus*, *Rhodopseudomonas*, *Corynebacterium*, *Rhodopseudomonas*, *Microbacterium*, *Achromobacter*, *Methylobacter*, *Methylosinus*, and *Methylocystis*. Preferred genera include *Bacillus*, *Pseudomonas*, *Protaminobacter*, *Micrococcus*, *Arthrobacter* and/or *Corynebacterium*. Examples of suitable species include, but are not limited to: *Bacillus subtilis*, *Bacillus cereus*, *Bacillus aureus*, *Bacillus acidi*, *Bacillus urici*, *Bacillus coagulans*, *Bacillus mycoides*, *Bacillus circulans*, *Bacillus megaterium*, *Bacillus licheniformis*, *Pseudomonas ligustri*, *Pseudomonas orvilla*, *Pseudomonas methanica*, *Pseudomonas fluorescens*, *Pseudomonas aeruginosa*, *Pseudomonas oleovorans*, *Pseudomonas putida*, *Pseudomonas boreopolis*, *Pseudomonas pyocyanea*, *Pseudomonas methylphilus*, *Pseudomonas brevis*, *Pseudomonas acidovorans*, *Pseudomonas methanoloxydans*, *Pseudomonas aerogenes*, *Protaminobacter ruber*, *Corynebacterium simplex*, *Corynebacterium hydrocarboxydans*, *Corynebacterium alkanum*, *Corynebacterium oleophilus*, *Corynebacterium hydrocarboclastus*, *Corynebacterium glutamicum*, *Corynebacterium viscosus*, *Corynebacterium dioxydans*, *Corynebacterium alkanum*, *Micrococcus cerificans*, *Micrococcus rhodius*, *Arthrobacter rufescens*, *Arthrobacter parafficum*, *Arthrobacter citreus*, *Methanomonas methanica*, *Methanomonas methanooxydans*, *Methylomonas agile*, *Methylomonas albus*, *Methylomonas rubrum*, *Methylomonas methanolica*, *Mycobacterium rhodochrous*, *Mycobacterium phlei*, *Mycobacterium brevicale*, *Nocardia salmonicolor*, *Nocardia minimus*, *Nocardia corallina*, *Nocardia butanica*, *Rhodopseudomonas capsulatus*, *Microbacterium ammoniaphilum*, *Achromobacter coagulans*, *Brevibacterium butanicum*, *Brevibacterium roseum*, *Brevibacterium flavum*, *Brevibacterium lacto-*

fermentum, *Brevibacterium paraffinolyticum*, *Brevibacterium ketoglutamicum*, and/or *Brevibacterium insectiphilum*.

[0126] In several embodiments, more than one type or species of microorganism is used. For example, in some embodiments, both algae and bacteria are used. In some embodiments, several species of yeast, algae, fungi, and/or bacteria are used. In some embodiments, a single yeast, algae, fungi, and/or bacteria species is used. In some embodiments, a consortium of cyanobacteria is used. In some embodiments, a consortium of methanotrophic microorganisms is used. In still additional embodiments, a consortium of both methanotrophic bacteria and cyanobacteria are used. In several embodiments, methanotrophic, heterotrophic, methanogenic, and/or autotrophic microorganisms are used.

[0127] In several embodiments of the invention, the microorganism culture comprises a consortium of methanotrophic, autotrophic, and/or heterotrophic microorganisms, wherein methane and/or carbon dioxide is individually, interchangeably, or simultaneously utilized for the production of biomass. In some embodiments, PHA-reduced biomass is used as a source of carbon by heterotrophic, autotrophic, and/or methanotrophic microorganisms. In several embodiments of the invention, the microorganism culture comprises methanotrophic microorganisms, cyanobacteria, and non-methanotrophic heterotrophic microorganisms, wherein methane and carbon dioxide are continuously utilized as sources of carbon for the production of biomass and PHA.

[0128] In some embodiments, microorganisms are employed in a non-sterile, open, and/or mixed environment. In other embodiments, microorganisms are employed in a sterile and/or controlled environment.

[0129] The terms “PHA”, “PHAs”, and “polyhydroxyalkanoate”, as used herein, shall be given their ordinary meaning and shall include, but not be limited to, polymers generated by microorganisms as energy and/or carbon storage vehicles; biodegradable and biocompatible polymers that can be used as alternatives to petrochemical-based plastics such as polypropylene, polyethylene, and polystyrene; polymers produced by bacterial fermentation of sugars, lipids, or gases; and/or thermoplastic or elastomeric materials derived from microorganisms. PHAs include, but are not limited to, polyhydroxybutyrate (PHB), polyhydroxyvalerate (PHV), polyhydroxybutyrate-covalerate (PHB/V), and polyhydroxyhexanoate (PHHx), as well as both short chain length (SCL), medium chain length (MCL), and long chain length (ICL) PHAs.

[0130] The terms “growth-culture medium”, “growth medium”, “growth-culture media”, “medium”, and “media”, as used herein shall be given their ordinary meaning and shall also refer to materials affecting the growth, metabolism, PHA synthesis, and/or reproductive activities of microorganisms. Non-limiting examples of growth-culture media used in several embodiments include a mineral salts medium, which may comprise water, nitrogen, vitamins, iron, phosphorus, magnesium, and various other nutrients suitable to effect, support, alter, modify, control, constrain, and/or otherwise influence the metabolism and metabolic orientation of microorganisms. A growth-culture medium may comprise water filled with a range of mineral salts. For example, each liter of a liquid growth-culture medium may be comprised of about 0.7-1.5 g KH_2PO_4 , 0.7-1.5 g

K_2HPO_4 , 0.7-1.5 g KNO_3 , 0.7-1.5 g NaCl, 0.1-0.3 g $MgSO_4$, 24-28 mg $CaCl_2 \cdot 2H_2O$, 5.0-5.4 mg EDTA $Na_4(H_2O)_2$, 1.3-1.7 mg $FeCl_2 \cdot 4H_2O$, 0.10-0.14 mg $CoCl_2 \cdot 6H_2O$, 0.08-1.12 mg $MnCl_2 \cdot 2H_2O$, 0.06-0.08 mg $ZnCl_2$, 0.05-0.07 mg H_3BO_3 , 0.023-0.027 mg $NiCl_2 \cdot 6H_2O$, 0.023-0.027 mg $NaMoO_4 \cdot 2H_2O$, and 0.011-0.019 mg $CuCl_2 \cdot 2H_2O$. A growth-culture medium can be of any form, including a liquid, semi-liquid, gelatinous, gaseous, foam, or solid substrate.

[0131] In several embodiments of the invention, a microorganism culture is produced in a liquid growth medium, wherein carbon dioxide and methane are utilized as a gaseous source of carbon for the production of methanotrophic and/or autotrophic biomass. In some embodiments, PHA-reduced methanotrophic and/or PHA-reduced autotrophic biomass is utilized as a source of carbon for the production of heterotrophic biomass and heterotrophically-produced PHA. In some embodiments, the growth medium is manipulated to effect the growth, reproduction, and PHA synthesis of the microorganism culture. Methods for the production of methanotrophic microorganisms are disclosed in the art, and are described by Herrema, et al., in U.S. Pat. No. 7,579,176, which is hereby incorporated by reference in its entirety. Methods for the production of cyanobacteria are described by Lee, et al. ("High-density algal photobioreactors using light-emitting diodes," *Biotechnology and Bioengineering*, Vol. 44, Issue 10, pp. 1161-1167), which is hereby incorporated by reference in its entirety. Methods for the production of methane from biomass are described by Deublein, et al. ("Biogas from Waste and Renewable Resources, WILEY-VCH Verlag GmbH & Co. KGaA, 2008), which is hereby incorporated by reference in its entirety. In some embodiments, PHA synthesis may be effected through the manipulation of one of more elements of the culture medium, including through the reduction, increase, or relative change in either the total or bioavailable concentration of one or more of the following elements: nitrogen, phosphorus, oxygen, methane, carbon dioxide, magnesium, potassium, iron, copper, sulfate, manganese, calcium, chlorine, boron, zinc, aluminum, nickel, and/or sodium. Methods for the production of PHA are described by Herrema, et al., in U.S. Pat. No. 7,579,176.

[0132] In several embodiments of the invention, methanol is added to a culture of methanotrophic microorganisms utilizing a closed loop recycling gas stream comprising methane. In some embodiments, methanotrophic microorganisms are enabled to grow under conditions of, and consume, very low concentrations of methane by co-utilizing methanol as a carbon substrate. In the past, the growth of methanotrophic microorganisms was significantly reduced under low methane concentrations due to, among other things, low mass transfer rates. In some embodiments, by the addition of methanol in a closed loop gas recycling system, it is possible to effect the substantially complete elimination of methane by methanotrophic microorganisms.

[0133] In several embodiments of the invention, the diffusion of light is increased in a liquid growth culture media by reducing the density of the liquid in a light path. In some embodiments the culture comprises autotrophic microorganisms. In some embodiments, the application of gas bubbles into the media decreases the relative solids density of the light path, thus enabling an increased diffusion of light into a liquid culture media from a given light intensity energy.

[0134] In several embodiments of the invention, a series of submerged light rods are placed into a liquid culture to manipulate or adjust the culture conditions. In some embodiments, the culture comprises autotrophic microorganisms. In some embodiments, the light rods function to diffuse light, diffuse gas, act as static or dynamic mixers, assist in the circulation of a liquid culture media, and/or facilitate heat exchange through the circulation of a gas, liquid, and/or combination thereof.

[0135] Traditionally, pH control in a microorganism growth system is difficult and/or costly to maintain. In some embodiments, pH is controlled by varying the nitrogen source supplied to a microorganism growth system between pH-increasing and pH-reducing nitrogen sources, e.g., NO_3^- and NH_3^+ , respectively. In some embodiments, nitrogen sources are utilized that do not significantly affect the pH of the system, including, when applicable, complex nitrogen sources such as biomass. In additional embodiments, a closed loop system is employed to reduce changes in pH. In some embodiments, respiration-generated carbon dioxide counterbalances increases in pH caused by the utilization of pH-increasing nitrogen sources, such as nitrates.

[0136] A number of methods are known for the induction of gas into liquid, including static mixing, ejector mixing, propeller mixing, and/or a combination thereof. Simultaneously, it is also known that shear can be highly detrimental to microorganism growth, and can often impede or permanently deactivate microorganism metabolism. Thus, mass transfer in a gas-based system is often limited by the need to counterbalance sufficient mixing with shear considerations. In several embodiments of the invention, a vessel comprising liquid culture media is mixed with a gas, e.g. methane, under relatively high shear conditions, and then subsequently transferred to a vessel comprising liquid culture media maintained under relatively low shear conditions. In some embodiments, microorganism growth is primarily induced in the low shear vessel. In some embodiments, high gas transfer rates are effected in the first high shear vessel by mixing while performed in the low shear vessel by gaseous diffusion.

[0137] In another embodiment, a closed loop gas recycling system is maintained, wherein a vessel comprising gas-utilizing microorganisms is supplied with gas, wherein the gas is utilized by gas-utilizing microorganisms, and wherein the rate at which gas is added to the system is determined by the rate at which the pressure in the vessel changes in accordance with the conversion of gases into metabolic derivatives (such as biomass, carbon dioxide, and water). For example, a vessel containing methane-utilizing microorganisms may be pressurized to 60 psi with a combination of methane and oxygen; as the pressure in the system drops in accordance with the metabolism of the methane-utilizing microorganisms, additional methane and oxygen is added to the system such that the pressure of the vessel remains at 60 psi. It shall be appreciated that, in certain embodiments, higher or lower pressures are maintained. In some embodiments, the system is periodically flushed to remove carbon dioxide. In some embodiments, autotrophic microorganisms and a light injection system may be added to the system in order to convert carbon dioxide into additional oxygen, thereby substantially reducing or eliminating the need to flush the system and/or introduce oxygen.

[0138] In several embodiments, PHA synthesis is induced in a microorganism culture comprising methane-utilizing,

heterotrophic, and/or carbon dioxide-utilizing microorganisms wherein a PHA inclusion concentration (by dry biomass weight) is generated of between about 0.01% and 95%. In some embodiments, the inclusion concentration is between 25% and 80%, including 25-35%, 35% to 50%, 50% to 65%, 65% to 80%, and overlapping ranges thereof. In some embodiments, the inclusion concentration is between 0.01% and 55%, including, 0.01% to 1%, 1% to 5%, 5% to 10%, 10% to 15%, 15% to 20%, 20% to 25%, 25% to 30%, 30% to 35%, 35% to 40%, 40% to 45%, 45% to 50%, 50% to 55%, and overlapping ranges thereof. In some embodiments, PHA synthesis is induced in a methanotrophic, heterotrophic, and/or autotrophic microorganism culture wherein a PHA inclusion concentration (by dry biomass weight) is generated of between 20% and 80%, between 30% and 70%, between 40% and 60%, between 50% and 70%, including 50% to 55%, 55 to 60%, 60% to 65%, 65% to 70%, and overlapping ranges thereof. In some embodiments of the invention, PHA synthesis is induced in microorganism culture comprising methanotrophic, autotrophic, and heterotrophic microorganisms, wherein an average PHA inclusion concentration (by dry biomass weight) is greater than 5%, greater than 20%, greater than 40%, greater than 65%, or greater than 70% by dry cell weight.

[0139] In some embodiments, the growth culture media is manipulated to induce both i) microorganism growth and ii) PHA synthesis within one or more open, non-sterile, or sterile vessels using a feast-famine culture regime. In some embodiments, the microorganisms are subject to successive alternating periods of nutrient/carbon availability and nutrient/carbon unavailability to encourage the reproductive success of microorganisms that are capable of synthesizing PHA, particularly at high inclusion concentrations. Feast-famine regimes useful for the selection of PHA producing microorganisms, including PHA-producing methanotrophic microorganisms, over microorganisms that either cannot produce PHA, produce PHA slowly, or produce PHA at relatively low concentrations are described in the art (Frigon, et al., “rRNA and Poly-Hydroxybutyrate Dynamics in Bioreactors Subjected to Feast and Famine Cycles,” *Applied and Environmental Microbiology*, April 2006, p. 2322-2330; Müller, et al., “Adaptive responses of *Ralstonia eutropha* to feast and famine conditions analysed by flow cytometry,” *J Biotechnol.* 1999 Oct. 8; 75(2-3):81-97; Reis, et al., “Production of polyhydroxyalkanoates by mixed microbial cultures,” *Bioprocess and Biosystems Engineering*, Volume 25, Number 6, 377-385, DOI: 10.1007/s00449-003-0322-4.)

[0140] In some embodiments of the invention, the classical feast-famine regime is modified to reduce PHA losses. Specifically, in the past, feast-famine regimes were thought to be effective by passing a microorganism culture through a period wherein carbon or nutrients were unavailable or relatively limited for metabolism, thereby forcing the culture to accumulate and/or consume intracellular PHA as a source of carbon to survive, and thereby selecting for microorganisms with the capacity to synthesize and store PHA. Applicant has surprisingly discovered that, some microorganisms with higher concentrations of intracellular PHA reproduce more efficiently than microorganisms with lower concentrations of intracellular PHA in periods of carbon availability and nutrient balance. Thus, in one embodiment, a novel PHA production regime is employed in one or more vessel wherein microorganisms are subjected to two successive and

recurring phases: 1) growth, wherein carbon and nutrient availability is optimized for reproduction, and 2) PHA synthesis, wherein carbon is available in excess, and one or more nutrient is reduced or increased relative to the growth period to induce PHA synthesis. In some embodiments, a fraction of the vessel media is removed for downstream PHA extraction and processing after the PHA synthesis period, and that fraction is replaced with lower cell density media, which simultaneously returns carbon and nutrient concentrations to reproductively favorable levels, e.g., the growth phase or growth conditions, and causes microorganisms to enter into a reproductive phase without consuming significant portions of intracellular PHA. As such, efficient PHA producing microorganisms selectively reproduce over inefficient or non-PHA producing microorganisms. As a result, some embodiments, i) increase the speed of the microorganism selection process by removing the PHA consumption step typical to previous feast famine models and ii) reduce the loss of PHA to cellular metabolism. According to such embodiments, the feast famine model is converted to a feast-polymerization-feast process. In several embodiments methanotrophic, autotrophic, and/or heterotrophic cultures are used in the feast-polymerization-feast process.

Removing a Portion of the PHA-Containing Biomass from the Culture, and Extracting PHA from the Removed PHA-Containing Biomass to Produce Isolated PHA and PHA-Reduced Biomass

[0141] In several embodiments, following the production of a microorganism culture comprising biomass and PHA (discussed above), at least a portion of the PHA-containing biomass is removed from the culture. In several embodiments, a portion ranging from 20% to 80% of the PHA-containing biomass is removed, including 30%-70%, 40% to 60%, 45% to 55%, and overlapping ranges thereof. Removal of PHA-containing biomass may be performed by a number of methods, including centrifugation, filtration, density separation, flocculation, agglomeration, spray drying, or other separation technique. In some embodiments, dewatering (e.g., by centrifugation) results in a biomass having a desirable water content that facilitates downstream processing of the biomass. For example, in some embodiments, centrifugation of the PHA-containing biomass reduces the amount of culture media (increases the relative biomass concentration) to a concentration range between about 100 and 500 grams of biomass per liter of culture media. In some embodiments, the concentration of the biomass is adjusted to about 100 to 200 g/L, 200 to 300 g/L, 300 to 400 g/L, 400 to 500 g/L, and overlapping ranges thereof. Advantageously, such an approach also produces, as an effective by-product, clarified culture media that can be optionally treated, measured, or recycled into one or more culture vessels.

[0142] In some embodiments, after a portion of the PHA-containing biomass is removed from the culture, PHA is extracted from the removed PHA-containing biomass to produce isolated PHA and PHA-reduced biomass.

[0143] As used herein, the terms “extraction” and “PHA extraction” shall be given their ordinary meaning and shall be used interchangeably to describe the removal and/or separation of PHA from biomass. PHAs may be extracted from biomass by several processes, including, but not limited to, the use of chemicals, mechanical means, solvents, and enzymes. These processes include the use of: i) solvents, such as acetone, ethanol, methanol, methylene chloride, and

dichloroethane, with and/or without the application of pressure and/or elevated temperatures, ii) supercritical carbon dioxide, iii) enzymes, such as protease, iv) surfactants, v) pH adjustment, including the protonic or hydroxide-based dissolution of non-PHA biomass, and/or vi) hypochlorite to dissolve non-PHA biomass, including the use of hypochlorite in conjunction with a solvent, such as methylene chloride. In some embodiments of the invention, PHA is extracted by solvent extraction from a PHA-containing biomass comprising gas-utilizing microorganisms and/or biomass-utilizing microorganisms to produce isolated PHA and PHA-reduced biomass. In some solvent-based embodiments, solvents suitable for dissolving the PHA are used, including carbon dioxide, acetone, methylene chloride, chloroform, water, ethanol, and methanol. In some embodiments, particular ratios of solvent to PHA provide optimal dissolution of PHA from the culture, and therefore lead to improved extraction and isolation efficiency and yield. For example, in some embodiments, ratios of solvent to PHA (in grams) of about 500:1 are used. In some embodiments, ratios of about 0.01:1 are used. In some embodiments, ratios ranging from between about 500:1 and 0.01:1 are used, such as 0.05:1, 1.0:1, 1.5:1, 20:1, 250:1, 300:1, 350:1, 400:1, or 450:1.

[0144] As discussed above, changes in temperature and/or pressure may also be used to facilitate the extraction of PHA from the PHA-containing biomass. In some embodiments, the extraction solvent chosen dictates the limits of temperature, pressure, and/or incubation times that are used. In some embodiments, solvent is combined with PHA-containing biomass and incubated for several minutes up to several hours. For example, in some embodiments, incubation is for about 10 minutes, while in other embodiments, overnight incubation times are used. In some embodiments, incubation times range from 30 minutes to about 1 hour, about 1 hour to about 2 hours, about 2 hours to about 4 hours, about 4 hours to about 6 hours, about 6 hours to about 8 hours, about 8 hours to about 10 hours, and from about 10 hours to overnight. Choice of incubation time is determined by solvent, culture density (e.g., number of microorganisms), type of organisms, expected PHA yield, and other similar factors.

[0145] Incubation temperature is also tailored to the characteristics of a given culture. Incubation temperatures can range from below room temperature to elevated temperatures of up to about 150° C. or about 200° C. For example, depending the solvent and other variables of the culture, temperatures are used that range from about 10° C. to 25° C., from about 25° C. to about 40° C., from about 40° C. to about 55° C., from about 55° C. to about 60° C., from about 60° C. to about 75° C., from about 75° C. to about 90° C., from about 90° C. to about 105° C., from about 105° C. to about 120° C., from about 120° C. to about 135° C., from about 135° C. to about 150° C., from about 150° C. to about 200° C., and overlapping ranges thereof.

[0146] As can be appreciated, if changes in temperature are made to a culture in a closed vessel, changes in pressure result. In some embodiments, increased pressure provides a shearing effect that aids in the liberation of PHA from the microorganisms. In some embodiments, pressure is regulated within a particular range. For example, in some embodiments, pressure of the reaction of the PHA-containing biomass with solvent occurs between about 40 and 200 psi, including about 50 to 60 psi, 60 to 70 psi, 70 to 80 psi,

80 to 90 psi, 90 to 100 psi, 100 to 125 psi, 125 to 150 psi, 150 to 175 psi, 175 to 200 psi and overlapping ranges thereof. Additional sources of shear (e.g., agitation, pumping, stirring etc.) are optionally used in some embodiments to enhance the extraction of PHA. Any one, or combination, of the PHA extraction methods described herein, or disclosed in the art, may be utilized as a method to carry out PHA extraction and remove PHA from the PHA-containing biomass.

[0147] In several embodiments, a solvent-based extraction system is utilized to carry out PHA extraction. In some embodiments, solvents are utilized to carry out PHA extraction at high temperatures, wherein PHA extraction occurs simultaneous with a temperature-enhanced breakdown or dissolution of PHA-containing biomass. In some embodiments, one or more solvent is utilized that is biodegradable and metabolically assimilable by the culture, such that PHA-reduced biomass comprising biomass and one or more biodegradable solvent may be contacted with the culture, and both the PHA-reduced biomass and the solvent may be utilized by the culture as a source of carbon. Non-limiting examples of such solvents include carbon dioxide, acetone, ethanol, methanol, and methylene chloride, among others.

[0148] In several embodiments a mixture of solvent and PHA comprises multiple phases, e.g. an aqueous phase and an organic phase. In some embodiments, solvent-based extraction comprises a more uniform mixture of solvent and PHA. In some embodiments, depending on the solvent the phases are separated prior to recovery of the PHA. In some embodiments, centrifugation is employed to further distinguish and separate the phases of the mixture (e.g., separation of the solvent-PHA phase from the water-biomass phase). In some embodiments, heat is also employed to maintain the solubility of the PHA in a given solvent

[0149] It shall be appreciated that the solubility of PHA varies with the solvent used, and therefore the temperature (if adjusted) and the separation techniques are tailored to match the characteristics of a given solvent. Thus, in some embodiments employing centrifugation, for example, a low speed centrifugation is used to separate the solvent-PHA phase from the water-biomass phase. In other embodiments, depending on the solvent, higher speed centrifugation is used. In some embodiments, centrifugation is employed in stages, e.g., low speed centrifugation followed by high speed centrifugation. Any of a variety of centrifuges can be employed, depending on the solvent used, for example, basket centrifuges, swinging bucket centrifuges, fixed rotor centrifuges, disc-back centrifuges, supercentrifuges, or ultracentrifuges.

[0150] In some embodiments, adjustable discharge ports suitable for a particular centrifuge are used in order to control the rate and degree of separation of solvent-PHA phase from the water-biomass phase. In some embodiments, the concentration of water in the water-biomass phase is adjusted to allow for suitable flow of the mixture through the centrifuge (or within a centrifuge tube). For example, in some embodiments, flow is suitable for separating the phases when the concentration of biomass (relative to water) is between about 1 and 100 g/L. In some embodiments, the concentration is between about 10 to 20 g/L, 20 to 30 g/L, 30 to 40 g/L, 40 to 50 g/L, 50 to 60 g/L, 60 to 70 g/L, 70 to 80 g/L, 80 to 90 g/L, 90 to 100 g/L, 100 to 200 g/L, 200 to 400 g/L, 400 to 600 g/L, and overlapping ranges thereof.

[0151] In still additional embodiments, increases in temperature not only facilitate the extraction of the PHA, they also facilitate the isolation of the PHA from the solvent (e.g. increased temperature increases solvent evaporation).

[0152] In some embodiments, an extraction process is carried out to remove PHA from a microorganism in such a manner that the microorganism is deactivated. In some embodiments, the deactivation is permanent, while in some embodiments the deactivation is temporary. Without being bound by theory, it is believed that PHA extraction techniques which do not permanently disable microorganisms enable the PHA-reduced biomass generated thereby to contribute to the metabolism of carbon sources after a PHA extraction process, including through intracellular and extracellular metabolism. In one embodiment, methods useful for the temporary disablement of microorganisms include solvent extraction, including solvent extraction carried out below 100° C., and particularly at intracellular temperatures below 100° C., including extraction temperatures of about 10° C. to 30° C. to 50° C. to 60° C., 60° C. to 70° C., 70° C. to 80° C., 80° C. to 90° C., 90° C. to 100° C., and overlapping ranges thereof.

[0153] In several embodiments, the PHA concentration of PHA-containing biomass is reduced as a result of the PHA extraction process. In several embodiments, the PHA concentration of PHA-containing biomass is reduced by at least 0.01% (by dry cell weight). In some embodiments, the PHA concentration is reduced by about 10%-50%, 50% to 70%, 70% to 75%, 75% to 80%, 80% to 85%, 85%, to 90%, 90% to 95%, 95%-99.9%, and overlapping ranges thereof.

[0154] While a variety of methods are known to enable the extraction PHA from biomass, most methods can be categorized into one of two classes: a) solvent-based extraction, or b) NPCM (non-polymer cellular material) dissolution-based extraction. NPCM dissolution-based extraction methods utilize chemicals (such as hypochlorite, or bleach), enzymes (such as protease), heat (especially to reach temperatures above 100° C.), pH (acids and bases), and/or mechanical means (such as homogenization) to break down, oxidize, and/or emulsify non-PHA cellular material. In some cases, extraction methods from both categories can be combined, such as the simultaneous utilization of hypochlorite and methylene chloride.

[0155] NPCM dissolution-based extraction methods require continuous and non-recoverable chemical inputs, such as hypochlorite, peroxides, enzymes, and pH adjustors, and also generate significant waste disposal issues. Thus, while these methods are effective, the use of solvent-based extraction methods is generally preferred in the industry due to the capacity of solvents to be distilled and recovered for continuous re-utilization in a closed loop cycle. Unfortunately, despite its benefits, some solvent-based extraction methods are energy intensive processes that play a major role in the high cost of PHA production, often accounting for more than 50% of total production costs. Accordingly, there exists a significant need for a novel method to significantly increase the energy efficiency of solvent-based extraction.

[0156] In several embodiments, a process for the extraction of polyhydroxyalkanoates from biomass using a solvent-based extraction method is provided., wherein the energy required to carry out the process is reduced relative to prior solvent-based extraction methods. Specifically, in one embodiment a high efficiency PHA extraction process is provided comprising providing a PHA-containing biomass

comprising PHA and water, mixing the biomass with a solvent at a temperature sufficient to dissolve at least a portion of the PHA into the solvent and at a pressure sufficient to enable substantially all or part of the solvent to remain in liquid phase, thereby producing a PHA-lean biomass phase and a PHA-rich solvent phase comprising solvent, water, and PHA, separating the PHA-rich solvent phase from the PHA-lean biomass phase at a temperature and pressure sufficient to enable substantially all or part of the solvent to remain in the liquid phase and prevent substantially all or part of the PHA within the PHA-rich solvent phase from precipitating, reducing the pressure or increasing the temperature of the PHA-rich solvent phase to cause the solvent to vaporize and the PHA to precipitate or become a solid while maintaining the temperature and/or the pressure of the PHA-rich solvent phase to prevent all or part of the temperature-dependent precipitation of the PHA into water, and collecting the solid PHA material, including optionally separating the precipitated PHA from the solvent and/or the water.

[0157] In the past, PHA precipitation has been induced in PHA-rich solvent by a) adding a non-PHA solvent to the solvent phase to reduce the solubility of PHA in the solvent phase and/or b) reducing the temperature of the solvent phase to reduce the solubility of PHA in the solvent. In particular, some methods 1) dissolve PHA in a solvent by increasing the temperature of the solvent and 2) precipitate PHA by reducing the temperature (and, thus, solubility) of the solvent. Other methods require adding water to a solution of PHA-rich solvent comprising dissolved PHA, wherein the addition of water to the solution reduces the solubility of the PHA in the solvent and causes the PHA to precipitate into the solvent and/or water. (For example, U.S. Pat. Nos. 4,562,245; 4,968,611; 5,894,062; 4,101,533, all herein incorporated by reference.) In each of these cases, energy efficiency is compromised; specifically, by adding water or a non-PHA solvent to reduce the PHA solubility of a solvent, additional energy is required for downstream water/non-solvent removal, heating, and/or distillation. By reducing the temperature of the solvent to reduce the solubility of the solvent and induce PHA precipitation, heat energy is redundantly expended, as the solvent must be re-heated for distillation and recovery. Therefore, in several embodiments, rather than adding a non-solvent to a PHA-solvent or reducing the temperature of the PHA-solvent to effect PHA precipitation, pressure and/or an increase in temperature is used to induce the precipitation or solidification of the PHA without redundantly reducing the temperature of solvent. Thus, in such embodiments, there is a significant reduction in the energy required to heat and/or distill non-solvent and/or solvent in downstream PHA processing

[0158] The removal of non-PHA materials from PHA often accounts for a significant fraction of PHA production costs. As a specific example, pigments often cause unwanted discoloration of PHA, and must be removed through costly processes, such as ozonation, peroxide washing, acetone washing, ethanol washing, solvent refluxing, hypochlorite digestion, enzymatic degradation, surfactant dissolution, or other methods disclosed in the art. In several embodiments, a non-sterile process is used to select for microorganisms exhibiting minimal pigmentation. Applicant has surprisingly discovered that, by manipulating the concentration of dissolved oxygen in a microorganism system, a culture may be

selected wherein white, tan, off-white, light brown, and/or light yellow pigments are exhibited rather than purple, red, pink, dark brown, orange, or other heavy pigments. Specifically, in some embodiments, an excess of dissolved oxygen is introduced into a growth media over successive periods, resulting in selective growth of strains of methanotrophic microorganisms which produce white, tan, off-white, light brown, and light yellow pigments, rather than those producing pink, red, purple, dark yellow, dark brown, and/or other heavy pigments. As a result, such embodiments, reduce the need for costly downstream pigmentation removal.

[0159] As used herein, the term “PHA-reduced biomass” or “substantially PHA-reduced biomass” shall be given its ordinary meaning and shall be used to describe a biomass material wherein the concentration of PHA relative to non-PHA material has been reduced in a PHA-containing biomass through the utilization of a PHA extraction process. In some embodiments, PHA-reduced biomass is further treated in a variety of ways. In some embodiments, the further treatment includes, but is not limited to, one or more of dewatering, chemical treatment, sonication, additional PHA extraction, homogenization, heat treatment, pH treatment, hypochlorite treatment, microwave treatment, microbiological treatment, including both aerobic and anaerobic digestion, solvent treatment, water washing, solvent washing, and/or drying, including simple or fractional distillation, spray drying, freeze drying, rotary drying, and/or oven drying.

[0160] In one embodiment, PHA-reduced biomass is substantially dried, wherein the resulting dried material comprises less than about 99% liquids, including water, solvents, salts, and/or growth-culture media. In some embodiments, the drying processes disclosed herein yield a dried material containing between about 95% and 75% liquids, between about 75% and 50% liquids, between about 50% and 25% liquids, between about 25% and 15% liquids, between about 15% and 10% liquids, between about 10% and 1% liquids, and overlapping ranges thereof. In some embodiments, drying is complete (e.g., between 1% 0.1% liquids, or less). In another embodiment of the invention, a liquid phase comprising PHA-reduced biomass is subjected to filtration, centrifugation, density differentiation, or other method to increase the solids content of the PHA-reduced biomass.

[0161] Traditionally, the separation of biomass from liquid growth media is difficult and impractical due to the plugging and fouling characteristics of biomass. In several embodiments, a method enabling the efficient filtration of microorganisms is provided. In some embodiments, a liquid chemical is added to the growth media comprising microorganisms, wherein the liquid chemical is ethanol, acetone, methanol, methylene chloride, ketones, alcohols, and/or chlorinated solvents, or a combination thereof. In some embodiments, microorganisms are then efficiently separated from liquid growth media using standard filtration equipment, such as a Buchner filter, filter press, or similar apparatus. In one embodiment, approximately 2 parts acetone are mixed with one part water, including both intracellular and extracellular water, to effect the efficient filtration of microorganisms comprising the water.

[0162] As used herein, the terms “isolated PHA” and “substantially isolated PHA” shall be given their ordinary meaning and shall refer to PHA that has been removed from a biomass material as a result of an extraction process, or a biomass material wherein the concentration of PHA relative

to non-PHA material has been increased by an extraction process. In several embodiments, isolated PHA is further treated in one or more of a variety of ways, including, but not limited to, purification, filtration, washing, oxidation, odor removal, pigment removal, lipid removal, non-PHA material removal, and/or drying, including centrifugation, filtration, spray drying, freeze drying, simple or fractional distillation, or density differentiation. Methods for the purification of PHA include the use of peroxides, water, hypochlorite, solvents, ketones, alcohols, and various other agents to separate and remove non-PHA material from PHA material. In some embodiments, PHA is removed from a microorganism culture by solvent extraction to produce isolated PHA in a PHA-rich solvent phase and PHA-reduced biomass in a PHA-lean liquid phase. In some embodiments, the solvent phase is separated from the liquid phase by filtration or centrifugation. In some embodiments, both centrifugation and filtration are used in combination (e.g., sequentially). In some embodiments, centrifugation is optionally followed by filtration. In other embodiments, filtration is optionally followed by centrifugation. Filtration, in some embodiments is performed under vacuum pressure, via gravity feed, under positive pressure, or in specialized filtration centrifuges. In some embodiments, the filter pore size is adjusted based on the species composition of the microorganism culture. In some embodiments, pore sizes of up to 200 μm are used. In some embodiments, smaller pore sizes are used, for example 15 to 20 μm , 10 to 15 μm , 5 to 10 μm , 1 to 5 μm , 0.001 to 1 μm , and overlapping ranges thereof.

[0163] In addition to the steps outlined above, additional steps may be taken to remove solvent from the extracted PHA, including evaporation, solvent casting, steam stripping, heat treatment, and vacuum treatment, each of which may be preferential, cost-effective, time-effective, or advantageous depending on the volatility and type of solvent used. In other embodiments, active processes can be used to reduce the solvent content of the solvent-PHA mixture. For example, in certain embodiments, alterations in temperature of certain solvents change the solubility of the PHA in the solvent, which effectively removes solvent from the PHA (e.g., the solvent is now separable from a precipitated PHA). In some embodiments, filtration, solvent temperatures, or vacuum treatment can be increased to reduce a portion of the solvent. In some embodiments, solvent to PHA ratios post extraction, filtration, evaporation, solvent casting, steam stripping, heat treatment, and/or vacuum treatment range from about 0.1:1 to about 1,000:1, including about 0.2:1, 0.3:1, 4.0:1, 5.0:1, 10.0:1, 20.0:1, 60:1, 70:1, 80:1, 90:1, 100:1, 200:1, 500:1, and 900:1.

[0164] As a result of the processes disclosed above, in some embodiments, the solvent is substantially removed from the isolated PHA in the PHA-rich solvent phase and the liquid is substantially removed from the PHA-reduced biomass in the PHA-lean liquid phase. In some embodiments, the isolated PHA is dried in a heated vessel to produce substantially pure isolated PHA (e.g., at least 80% PHA by dry weight, preferably at least 98% PHA by weight, more preferably at least 99% PHA by weight).

[0165] Numerous varieties of heated or drying vessels may be used to dry the isolated PHA, including ovens, centrifugal dryers, air dryers, spray dryers, and freeze dryers, among others. In some embodiments, heat is applied to a drying vessel to speed the process and/or to remove (e.g., evaporate traces of solvent from the PHA). It shall be

appreciated that the moisture content of the isolated PHA will depend, in some embodiments, on the solvent used, and the corresponding separation technique used (as described above). For example, a volatile solvent in combination with ultracentrifugation would result in a less moist extracted PHA, while a less active separation technique (e.g., gravity phase separation) would yield a more moist extracted PHA. In some embodiments, internal dryer temperatures range from 20° C. to 40° C. to about 200° C. In some embodiments, internal temperatures range from about 50° C. to 90° C., about 90° C. to 180° C., about 65° C. to 175° C. and overlapping ranges thereof. In some embodiments, outlet temperatures are substantially lower than inlet on internal temperatures. In some embodiments, outlet temperatures range from 30° C. to 90° C. In some embodiments, outlet temperatures are between about 35° C. to 40° C., about 40° C. to 45° C., about 45° C. to 50° C., about 50° C. to 55° C., about 55° C. to 90° C., and overlapping ranges thereof. It shall also be appreciated that the internal and outlet temperatures may be adjusted throughout the drying process (e.g., the temperature difference may initially be large, but decrease over time, or vice versa).

[0166] From the above discussion, it shall be appreciated that the type of dryer used, and the temperatures used (if other than atmospheric temperatures) are easily tailored to correspond to the techniques used in the extraction process. In some embodiments, particular dryer components are beneficial in the isolation of PHA. For example, depending on the moisture content of the extracted PHA (e.g., the amount of residual solvent) particular components of an evaporative-type dryer, such as an oven dryer, rotary dryer, spin flash dryer, spray dryer (equipped with various types of nozzle types, including rotary atomizer, single flow atomizer, mist atomizer, pressure atomizer, dual-flow atomizer) convection heat dryer, tray dryer, scrape-flash dryer, or other dryer type are used. By way of additional example, if a freeze dryer (e.g., a lyophilizer) is used, in some embodiments a manifold dryer is used, optionally in conjunction with a heat source. Also by way of example, a tray lyophilizer can be used, in some embodiments, with the isolated and dried PHA being stored and sealed in containers (e.g., vials) before re-exposure to the atmosphere. In certain embodiments, such an approach is used when long-term storage of the PHA is desired.

[0167] It shall also be appreciated that certain varieties of heated/drying apparatuses have adjustable flow rates that can be tailored to the moisture content of the isolated PHA. For example, an isolated PHA having a high moisture content would be fed into a dryer at a slower input rate to allow a higher degree of drying per unit of PHA inputted into the dryer. Conversely, a low moisture content isolated PHA would likely require less time to dry, and therefore could be input at a faster rate. In some embodiments, input rates of isolated PHA range from several hundred liters of isolated PHA-solvent mixture per minute down to several milliliters per minute. For example, input rates can range from about 10 mL/min to about 6 L/min, including about 10 ml/min to about 50 ml/min, about 50 ml/min to about 100 ml/min, about 100 ml/min to about 500 ml/min, about 500 ml/min to about 1 L/min, about 1 L/min to about 2 L/min, about 2 L/min to about 4 L/min, about 4 L/min to about 6 L/min, and about 100 L/min to about 500 L/min.

[0168] A range of PHA functional characteristics can be attained by mixing one PHA molecule, such as PHB, with

various PHA polymers, including PHB, at various molecular weights. Therefore, in several embodiments, a first isolated PHA is heated to reduce the molecular weight of the first isolated PHA, and then subsequently mixed with a second PHA wherein the molecular weight of the second PHA is higher than the molecular weight of the first PHA. With such embodiments, Applicant has surprisingly discovered methods to functionalize one or more PHAs, including PHB. In additional embodiments of the invention, the molecular weight of a first PHA is reduced from about 800,000-5,000,000 Daltons to about 30,000 to 800,000 Daltons and mixed with a second PHA with a molecular weight of about 800,000 to 5,000,000 Daltons to modify the functionalities of the input PHAs. In yet another embodiment, a first PHA is mixed with a second PHA wherein the molecular weight of the first PHA is at least 0.1% less than the molecular weight of the second PHA. In some embodiments, the difference in molecular weight between the first and second PHA is about 0.1% to 1%, about 1% to 10%, about 10% to 20%, about 20% to 30%, about 30% to 40%, about 40% to 50%, about 50% to 60%, about 60% to 70%, and overlapping ranges thereof. In still additional embodiments, PHAs having greater differences in molecular weight are used. In yet another embodiment, the molecular weight of a first PHB is reduced to less than about 100,000-500,000 Daltons and mixed with a second PHA with a molecular weight greater than about 100,000-500,000 Daltons to modify the functionality of the input PHB. It shall be appreciated that input PHA and PHB weight may vary from the ranges disclosed above, but based on the differences in the molecular weights, the alteration in functionality of the input PHB is still achieved.

Purifying the Isolated PHA

[0169] In some embodiments, isolated PHA is purified to produce a PHA material that is substantially pure PHA. In some embodiments, the isolated PHA is purified to at least 20% pure PHA by dry weight. In some embodiments, the isolated PHA is purified to at least 55% pure PHA by dry weight, while in some embodiments, the isolated PHA is purified to at least 90% pure PHA by dry weight. In additional embodiments, purity of the isolated PHA is between about 90 and 99.9%, including 91, 92, 93, 94, 95, 96, 97, 98, and 99% pure.

[0170] In several embodiments, isolated PHA may be recovered by any one, or a combination, of the methods described above, including, but not limited to: washing, filtration, centrifugation, dewatering, purification, oxidation, non-PHA material removal, solvent removal, and/or drying. In some embodiments, isolated PHA is recovered according to the manner in which it has been removed from the culture. For example, in embodiments in which solvent-based extraction is employed, a recovery method may be employed to remove the isolated PHA from the solvent and/or other non-PHA material. In one embodiment, solvent may be used to extract the PHA, wherein the solvent is then substantially removed from the isolated PHA by carrying out PHA precipitation and filtration, excess solvent distillation and/or removal, and/or drying, resulting in the recovery of dry, isolated PHA. In embodiments employing enzyme, surfactant, protonic, hydroxide, and hypochlorite-based extraction techniques wherein the dissolution of non-PHA material is induced, isolated PHA may be filtered, washed, separated, centrifuged, and/or dried, resulting in the recovery of dry, isolated, purified PHA. The resultant PHA, in

some embodiments, is further used in downstream processing, including thermoforming.

Returning PHA-Reduced Biomass to the PHA-Producing Culture to Convert PHA-Reduced Biomass into PHA

[0171] In several embodiments of the invention, the PHA-reduced biomass is returned to the culture to cause the biomass-utilizing microorganisms within the culture to convert the carbon within the PHA-reduced biomass into PHA. By using PHA-reduced biomass as a source of carbon for the production of microorganisms in a microorganism fermentation system, a series of biochemical enzymatic pathways are generated in situ by the microorganism culture to carry out the metabolic utilization of PHA-reduced biomass for growth, reproduction, and PHA synthesis.

[0172] Without being limited by theory, it is believed that gas-utilizing microorganisms and biomass-utilizing microorganisms are able to co-exist as a single microorganism consortium because they utilize sources of carbon that require distinctly different bioenzymatic assimilation pathways. For instance, while methane metabolism requires the methane monooxygenase enzyme to catalyze the conversion of methane into methanol for cellular assimilation, and methane monooxygenase is competitively inhibited by a wide range of compounds, it is not inherently deactivated by high concentrations of cellular biomass, including PHA-reduced biomass. Similarly, the chlorophyll-based metabolic assimilation systems required for the conversion of carbon dioxide into biomass and PHA are not inherently deactivated or competitively inhibited by high concentrations of cellular biomass, including PHA-reduced biomass. Likewise, the enzymatic architecture enabling the metabolic utilization, breakdown, and/or assimilation of PHA-reduced biomass is not inherently deactivated or competitively inhibited by high concentrations of methane and/or carbon dioxide, particularly as the process requires neither methane monooxygenase nor chlorophyll. Without being limited by theory, Applicant believes that the relatively non-competitive, and in some cases commensal or mutualistic relationships between microorganisms consuming a carbon-containing gas and a PHA-reduced biomass, make it possible to create a microorganism culture comprising biomass-utilizing microorganisms and gas-utilizing microorganisms, wherein both carbon-containing gases and PHA-reduced biomass may be metabolized as simultaneously assimilable sources of carbon.

[0173] In the case of autotrophic, methanotrophic, and/or biomass-utilizing microorganisms, Applicant has found that a mutualistic, positive-feedback loop relationship can be created in a single (or optionally multiple) reaction vessel. In such embodiments, the oxygen created by autotrophic metabolism is utilized by methanotrophic and/or biomass-utilizing microorganisms for metabolic functions, the carbon dioxide created by methanotrophic and/or biomass-utilizing microorganism metabolism is utilized for autotrophic metabolism, the methane and/or carbon dioxide created by anaerobic methanogenic microorganisms is utilized by methanotrophic microorganisms, and the biomass created by methanotrophic, autotrophic, and/or heterotrophic microorganisms is used to provide a source of carbon to methanogenic and/or other heterotrophic microorganisms. Due to the microscopic-level induction of oxygen and/or carbon dioxide created therein, mass transfer efficiencies in several embodiments are significantly improved over traditional gas induction means, such as gas sparging, mechanical mixing,

static mixing, or other means known in the art. To applicant's knowledge, prior to the disclosure herein, the use of autotrophic microorganisms cultured in association with heterotrophic microorganisms has never been suggested as a means to improve mass transfer efficiencies, supply oxygen, and/or augment microorganism growth rates in a positive feedback loop system.

[0174] In several further embodiments of the invention, PHA-reduced biomass is used by heterotrophic microorganisms, including acidogenic, acetogenic, and methanogenic microorganisms, to produce methane, which is further utilized by methanotrophic microorganisms to produce biomass, including PHA. In some embodiments of the invention, anaerobic microorganisms coexist with aerobic microorganisms under microaerobic conditions (e.g., mean dissolved oxygen concentrations approximately 0.00-1.0 ppm, including about 0.001 to 0.002 ppm, 0.002 to 0.03 ppm, 0.03 to 0.04 ppm, 0.04 to 0.5 ppm, 0.5 to 0.6 ppm, 0.6 to 0.7 ppm, 0.7 to 0.8 ppm, 0.8 to 0.9 ppm, 0.9 to 1.0 ppm, and overlapping ranges thereof.

[0175] In some embodiments of the invention, heterotrophic, methanotrophic, methanogenic, and/or autotrophic microorganisms are divided into multiple stages and vessels, in particular, anaerobic and aerobic stages, in order to carry out the conversion of PHA-reduced biomass into methane and PHA. In further embodiments of the invention, PHA-reduced biomass is returned to the culture using one or more vessels, whereby it is first converted to carbon dioxide, methane, and/or volatile organic compounds by a heterotrophic, e.g., methanogenic, microorganism consortium under anaerobic conditions and then converted to PHA by methanotrophic microorganisms under aerobic conditions, whereby carbon dioxide is also metabolized or otherwise used by autotrophic microorganisms, methanotrophic microorganisms, and heterotrophic microorganisms.

[0176] In several embodiments, light intensity is utilized to regulate the growth rate of heterotrophic and/or methanotrophic microorganisms. In some embodiments, light intensity is manipulated to regulate the generation of oxygen by autotrophic microorganisms. In some embodiments, the rate of oxygen generated by autotrophic microorganisms is subsequently used to control the growth and metabolism of heterotrophic and methanotrophic microorganisms.

[0177] In several embodiments, carbon dioxide is supplied to autotrophic microorganisms in the form of carbon dioxide created by methanotrophic and/or heterotrophic microorganisms. In some embodiments, each of these varieties of microorganism is cultured in a single vessel. In some embodiments, methane, sugar, biomass, and/or another non-carbon dioxide source of carbon is used to grow autotrophic microorganisms. To applicant's knowledge, autotrophic microorganisms have never been cultured using methane as a sole carbon input.

[0178] Some gas-utilizing microorganisms are unable to produce high concentrations of intracellular PHA. However, according to several embodiments, certain microorganism consortiums utilizing PHA-reduced biomass, or derivatives thereof, as a source of carbon are able to generate high intracellular PHA concentrations and thus effectively convert low PHA concentration biomass derived from a carbon-containing gas into a high PHA concentration biomass material under the conditions disclosed herein. In several embodiments, the concentration (by weight) of intracellular PHA is between about 10% to 30%, 30% to 50%, 50% to

70%, 70% to 80%, 80% to 90%, 90% or more, and overlapping ranges thereof. Thus, in one embodiment, the culture is contacted with the PHA-reduced biomass and then manipulated, according to the processes described herein, to effect PHA synthesis, wherein the PHA-reduced biomass is converted into PHA by biomass-utilizing microorganisms. In some embodiments, PHA synthesis is induced by nutrient limitation, nutrient excess, nutrient imbalance, or large shifts in nutrient concentration. In still further embodiments, PHA synthesis is induced by reducing the availability of nitrogen, oxygen, phosphorus, or magnesium to the culture. In some embodiments, these nutrients are simultaneously reduced (to varying or similar degrees). In some embodiments, the nutrients are reduced sequentially. In some embodiments, only one of the nutrients is reduced. For example, in certain embodiments, PHA synthesis is induced by reducing the availability of oxygen to the culture. In some embodiments, this is achieved by manipulating the flow rate of air or oxygen into the growth medium. In some embodiments, manipulation of the flow rate of other carbon-containing gases, such as methane and/or carbon dioxide, into the growth medium, or otherwise manipulating the rate of gas transfer in a system (e.g., by adjusting mixing rates or light injection rates) is employed. In one embodiment, oxygen limitation is induced by reducing the flow rate of oxygen into the growth medium. In another embodiment, oxygen limitation is induced by reducing the rate of light transmission into the medium to reduce the production of oxygen by autotrophic microorganisms. In some embodiments of the invention, the concentration of PHA generated in a biomass-utilizing microorganism culture utilizing PHA-reduced biomass as a source of carbon is at least 5%, at least 20%, or at least 50% by dry cell weight; in particularly preferred embodiments of the invention, the concentration of PHA in a biomass-utilizing microorganism is at least 80% by dry cell weight.

[0179] In some embodiments, a PHA-reduced biomass recycling system is utilized wherein substantially all (e.g., at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, or at least 98%) of the PHA-reduced biomass produced is contacted with the culture until it is converted into PHA. In some such embodiments, solid sources of carbon are substantially output from the process or culture only in the form of isolated PHA.

[0180] In several embodiments, as carbon-containing gases are continually added to the medium to effect the production of biomass, the process disclosed above is repeated. Specifically, as the process continues, a portion of the PHA-containing biomass from the culture is removed from the medium, PHA is extracted from the PHA-containing biomass, PHA-reduced biomass is separated from isolated PHA, isolated PHA is recovered, purified, and dried, and PHA-reduced biomass is sent back to the culture and converted by the culture into PHA. In one embodiment, substantially all PHA-reduced biomass produced is contacted with the culture until it is converted into isolated PHA, and solid sources of carbon are substantially output from the process only in the form of isolated PHA. In other embodiments, carbon is substantially output from the system only in the form of PHA and methane, carbon dioxide, and/or volatile organic compounds.

[0181] The following example is provided to further illustrate certain embodiments within the scope of the invention. The example is not to be construed as a limitation of any

embodiments, since numerous modifications and variations are possible without departing from the spirit and scope of the invention.

Example 1

[0182] A fermentation system comprising one or more vessels are partially filled with one or more liquid growth mediums, wherein the medium comprises methanotrophic, autotrophic, methanotrophic, and/or other heterotrophic or biomass-utilizing microorganisms containing PHA, and, per liter of water, 0.7-1.5 g KH_2PO_4 , 0.7-1.5 g K_2HPO_4 , 0.7-1.5 g KNO_3 , 0.7-1.5 g NaCl , 0.1-0.3 g MgSO_4 , 24-28 mg $\text{CaCl}_2 \cdot 2\text{H}_2\text{O}$, 5.0-5.4 mg $\text{EDTA Na}_4(\text{H}_2\text{O})_2$, 1.3-1.7 mg $\text{FeCl}_2 \cdot 4\text{H}_2\text{O}$, 0.10-0.14 mg $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$, 0.08-1.12 mg $\text{MnCl}_2 \cdot 2\text{H}_2\text{O}$, 0.06-0.08 mg ZnCl_2 , 0.05-0.07 mg H_3BO_3 , 0.023-0.027 mg $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$, 0.023-0.027 mg $\text{NaMoO}_4 \cdot 2\text{H}_2\text{O}$, 0.011-0.019 mg $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$. One or more of the mediums are anaerobic and/or aerobic, and carbon containing gases, including methane, carbon dioxide, and volatile organic compounds, as well as optionally air or oxygen, are fed into all or part of the system to induce the growth and reproduction of microorganisms through the utilization of carbon-containing gases, as well as the production of PHA.

[0183] Next, a portion of the media volume of the fermentation system is passed through a basket centrifuge to increase the solids content of the medium to approximately 167 g/L. The solids-containing concentrate phase of the centrifuged solution is then transferred to a PHA extraction vessel, and the substantially solids-free filtrate phase of the centrifuged solution is recycled back to the fermentation system.

[0184] In some embodiments, the solids-containing concentrate phase is optionally chemically pre-treated prior to extraction. In some embodiments, one or more of acids, bases, chloride, ozone, and hydrogen peroxide is added. In several embodiments, chemical pre-treatment increases the efficiency and yield of the subsequent extraction process. In some embodiments, the chemical pre-treatment functions to break down the cell wall (partially or fully), thereby liberating a greater portion of the intracellular PHA. In some embodiments, chemical pre-treatment dissolves and/or removes impurities that negatively impact the PHA extraction process. In some embodiments, chemical pre-treatment enhances cell agglomeration, which increases the percentage of microorganisms that are extracted (e.g., cells in an agglomerated mass are not separated or lost in transfer steps). Next, following optional chemical pre-treatment, solvent is added into the PHA extraction vessel to create a solvent solution, and the solvent solution is then mixed for a period of time to cause the PHA in both the microorganisms to dissolve into the solvent, and thereby create PHA-rich solvent and PHA-reduced biomass. Over the course of a defined mixing period (e.g., 0.1-10 hours), the PHA content of the microorganisms is reduced by about 80% as it is dissolved into the solvent.

[0185] Next, the solvent solution comprising the PHA-rich solvent and PHA-reduced biomass is passed through a filter located at the bottom of the PHA extraction vessel, and the PHA-rich solvent is thereby separated from the PHA-reduced biomass. Water is then added to the PHA extraction vessel to create a water-biomass solution, and the water-biomass solution is then heated to 75° C. to cause any remaining solvent associated with the PHA-reduced biomass to exit the PHA extraction vessel as a gaseous vapor.

The vapor discharged from the PHA extraction vessel is then passed through a heat exchanger and recovered as liquid solvent. Meanwhile, the PHA-rich solvent is transferred to a PHA purification vessel and mixed with room temperature water to create a water-solvent solution. The water-solvent solution is then heated to cause i) the solvent to exit the PHA extraction vessel as a gaseous vapor and if) the isolated PHA to precipitate into the water and/or become a solid. The solvent vapor created by heating the water-solvent solution is then passed through a heat exchanger and converted into liquid solvent.

[0186] The isolated PHA is then substantially dewatered by filtration in a Nutsche filter, and the Nutsche filter containing the substantially dewatered isolated PHA is then heated to remove any additional volatile compounds, including trace water and/or solvent. Following heat drying in the Nutsche filter, the isolated PHA is recovered as substantially pure PHA (e.g., greater than about 90% PHA).

[0187] Concurrently, the water-biomass solution comprising PHA-reduced biomass and water is transferred from the PHA extraction vessel back into the fermentation system, where the PHA-reduced biomass is contacted with one or more of the microorganism cultures. Next, the medium of the fermentation system is manipulated to cause the one or more microorganisms within the system to metabolize the PHA-reduced biomass as a source of carbon and nutrients.

[0188] Depending on the embodiment, the culture conditions are adjusted to determine the point at which the inception of the growth or PHA metabolism phase occurs. As discussed herein, manipulation of the concentration of one or more growth culture media nutrients can alter the metabolic pathways favored by certain microorganisms. Additionally, the use of PHA-reduced biomass-derived carbon for the production of additional biomass versus the production of PHA can be tailored based on whether the intent is to grow the culture (e.g., increase the overall biomass) or to harvest PHA (e.g., shift the culture from growth phase to production of PHA). As such, the reduction, increase, or adjustment of the concentration certain growth nutrients, and the timing of such adjustment, plays a role in the metabolic state and PHA production of the culture. Adjustment of growth nutrients can occur at any point after the PHA-reduced biomass is returned to the microorganism system. In some embodiments, adjustment is immediate (e.g., within minutes to a few hours). In some embodiments, a longer period of time elapses. In some embodiments, adjustment in one or more growth nutrients occurs after about 2 to 4 hours, after about 4 to 6 hours, after about 6 to 8 hours, after about 8 to 10 hours, after about 10 to 12 hours, after about 12 to 14 hours, after about 14 to 18 hours, after about 18 to 24 hours, and overlapping ranges thereof. In still additional embodiments, adjustment in one or more growth nutrients occurs after about 2 to 5 days, 5 to 10 days, 10 to 15 days, 15 to 20 days, 20 to 30 days, 30 to 50 days, and overlapping ranges thereof. In some embodiments, longer times elapse prior to adjusting one or more growth nutrients to induce PHA polymerization.

[0189] After a desired period of time has elapsed, the dissolved oxygen and/or nitrogen concentration (or concentration of another nutrient) of one or more parts of the medium is reduced or adjusted to cause one or more of the microorganisms within the system to utilize the PHA-reduced biomass in the medium as a source of carbon for the synthesis of PHA. In some embodiments, the percent adjust-

ment ranges from about 20% to about 100%, including 20% to 30%, 30% to 40%, 40% to 50%, 50% to 60%, 60% to 70%, 70% to 80%, 80% to 90%, 90% to 100%, and overlapping ranges thereof. It shall be appreciated that, depending on the characteristics of a culture in a given embodiment, a specific percentage reduction, increase, or adjustment in nutrient may not be necessary, but a reduction, increase, or adjustment is used that is sufficient to convert certain cells from a relative growth phase to a relative PHA synthesis phase. After approximately 12-24 hours of PHA synthesis, substantially all of the PHA-reduced biomass within the growth medium has been metabolized into biomass-utilizing microorganism biomass, including PHA. It shall be appreciated that, in certain embodiments, greater or lesser PHA synthesis times result in varying percentages of the PHA-reduced biomass within the growth medium being metabolized into biomass, including PHA.

[0190] As carbon containing gases are continually added to the fermentation system to effect the production of biomass, the process is repeated, wherein solid sources of carbon substantially exit the system only in the form of PHA. Specifically, as the process continues, a portion of the PHA-containing biomass from the fermentation vessel is passed through a dewatering centrifuge to increase the solids content of the PHA-containing biomass, PHA is extracted from the removed PHA-containing biomass using a solvent-based extraction system to create PHA-reduced biomass and isolated PHA, PHA-reduced biomass is separated from isolated PHA, isolated PHA is recovered, purified, and dried, and PHA-reduced biomass is sent back to the fermentation system and converted by microorganisms into PHA, such that substantially all PHA-reduced biomass produced is contacted with the culture until it is converted into isolated PHA, and wherein solid sources of carbon are substantially output from the process only in the form of isolated PHA.

[0191] While the above description of several compositions, systems, and methods contains many specificities, it should be understood that the embodiments of the present invention described above are illustrative only and are not intended to limit the scope of the invention. Numerous and various modifications can be made without departing from the spirit of the embodiments described herein. Accordingly, the scope of the invention should not be solely determined by the embodiments described herein, but also by the appended claims and their legal equivalents.

1. A process for the production of a polyhydroxyalkanoate (PHA), comprising the steps of:

- (a) providing a culture of microorganisms comprising microorganism-derived biomass, said microorganism-derived biomass comprising PHA,

wherein said culture of microorganisms can metabolize both a carbon containing gas and carbon within a PHA-reduced biomass to generate PHA;

- (b) removing a portion of said microorganism-derived biomass comprising PHA from said culture of microorganisms and subjecting said microorganism-derived biomass comprising PHA to a PHA removal process, wherein said removal process results in PHA-reduced biomass having at least 10% less PHA content as compared to a microorganism-derived biomass not subjected to the removal, wherein said PHA-reduced biomass comprises carbon;

- (c) increasing the concentration of PHA in said culture of microorganisms to produce PHA-rich biomass; and

(d) combining said PHA-reduced biomass with said PHA-rich biomass and returning the biomass combination to step (b),
 wherein the carbon within said PHA-reduced biomass is used by said microorganisms as a substrate to produce additional PHA.

2. The process of claim 1, wherein at least a portion of said carbon within said PHA-reduced biomass is non-PHA carbon.

3. The process of claim 1, wherein said carbon within said PHA-reduced biomass is metabolized by said PHA-rich biomass to produce converted biomass comprising carbon, wherein the carbon within said converted biomass is used by said PHA-rich biomass to produce PHA.

4. The process of claim 3, wherein said converted biomass is carbon dioxide and/or methane.

5. The process of claim 3, wherein said converted biomass is metabolically generated material, a volatile fatty acid, a volatile organic compound, a carbon-containing compound, intracellular, cellular, and/or extracellular material, a polymer, an amino acid, a nucleic acid, a carbohydrate, a lipid, a sugar, a polyhydroxyalkanoate, a chemical, and/or a metabolic derivative, intermediary, and/or end-product.

6. The process of claim 1, wherein said culture of microorganisms comprises a mixed culture of microorganisms comprising methanotrophic microorganisms, carbon-dioxide utilizing microorganisms, heterotrophic microorganisms, autotrophic microorganisms, cyanobacteria, biomass-utilizing microorganisms, methanogenic microorganisms, aerobic microorganisms, anaerobic microorganisms, acidogenic microorganisms, acetogenic microorganisms and combinations thereof.

7. The process of claim 1, wherein increasing the concentration of said PHA in said culture of microorganisms comprises reducing the concentration of water in said removed culture, wherein said PHA-rich biomass comprises said culture wherein the water concentration has been reduced, and wherein said PHA-reduced biomass comprises said culture, wherein the water concentration has not been reduced.

8. The process of claim 1, wherein increasing the concentration of PHA in said culture of microorganisms comprises extracting said PHA from said culture of microorganisms.

9. The process of claim 8, wherein extracting said PHA from said culture of microorganisms comprises mixing said culture of microorganisms with an extraction agent or mechanism selected from the group consisting of solvents, solvent washing, chemical treatment, microwave treatment, simple or fractional distillation, supercritical carbon dioxide, heat, enzymes, surfactants, acids, bases, hypochlorite, peroxides, polymers, bleaches, ozone, EDTA, and/or a combination thereof,
 wherein said extraction is optionally performed with a solvent selected from the group consisting of methylene chloride, acetone, ethanol, ketones, alcohol, chloroform, dichloroethane, water, and/or carbon dioxide, and
 wherein said extraction is optionally performed by mixing said removed PHA-containing biomass with a mechanism selected from the group consisting of sonication, homogenization, distillation, spray drying, hypochlo-

rite non-PHA dissolution, protonic non-PH dissolution, non-PHA dissolution, enzymatic treatment, and/or freeze drying.

10. The process of claim 1, wherein natural and/or artificial light is utilized to influence the metabolism of said culture.

11. The process of claim 1, wherein said culture uses at least a portion of a carbon-containing gas to produce said PHA,

wherein said gas is derived from one or more sources from the group consisting of: landfills, wastewater treatment plants, power production facilities or equipment, agricultural digesters, oil refineries, natural gas refineries, natural gas streams, cement production facilities, and/or anaerobic organic waste digesters.

12. The process of claim 1, wherein said culture uses at least a portion of a carbon-containing gas to produce said PHA, wherein said gas is derived from one or more sources from the group consisting of: landfill, wastewater treatment plant, power production facility, or anaerobic decomposition source.

13. The process of claim 1, wherein the metabolism, growth, reproduction, and/or PHA synthesis of said culture is controlled, manipulated, and/or affected by a culture medium,

wherein said culture medium comprises nutrients, and wherein said nutrients comprise at least one or more of the following: water, nitrogen, methane, carbon dioxide, phosphorus, oxygen, magnesium, potassium, iron, copper, sulfur, manganese, molybdenum, calcium, chlorine, boron, zinc, aluminum, nickel, and/or sodium.

14. The process of claim 13, wherein said production of said PHA is caused by increasing, reducing, or maintaining the concentration of one or more nutrient within said culture medium.

15. The process of claim 13, wherein said culture is caused to grow and/or reproduce by increasing, reducing, or maintaining the concentration of one or more nutrient within said medium.

16. The process of claim 13, wherein said culture medium is controlled to cause said culture of microorganisms to undergo a first period of growth and reproduction, wherein said culture medium is subsequently controlled to shift the culture of microorganisms to a period of PHA synthesis, wherein said medium is subsequently further controlled to shift the culture of microorganisms to an additional period of growth and reproduction, wherein microorganisms within said culture of microorganisms that reproduce more efficiently as a result of higher intracellular concentrations of PHA are produced in higher concentrations than microorganisms within said culture of microorganisms that reproduce less efficiently as a result of lower intracellular concentrations of PHA.

17. The process of claim 16, wherein said microorganisms that reproduce more efficiently as a result of higher intracellular concentrations of PHA reproduce more efficiently by using said intracellular PHA as a source of carbon, energy, power, or reducing power.

18. The process of claim 16, wherein the shift from said first period of growth and reproduction to said PHA synthesis period and the shift from said PHA synthesis period to said additional period of growth and reproduction is continuous for at least part of the culture.

19. A process for extracting PHA from a PHA-containing biomass, the process comprising:

- (a) providing a culture of microorganisms comprising microorganism-derived biomass, said microorganism-derived biomass comprising PHA, wherein said culture of microorganisms can metabolize both a carbon containing gas and carbon within a PHA-reduced biomass to generate PHA;
- (b) removing a portion of said microorganism-derived biomass comprising PHA from said culture of microorganisms and subjecting said microorganism-derived biomass comprising PHA to a PHA removal process, wherein said removal process results in PHA-reduced biomass having at least 10% less PHA content as compared to a microorganism-derived biomass not subjected to the removal, wherein said PHA-reduced biomass comprises carbon;
- (c) increasing the concentration of PHA in said culture of microorganisms to produce a PHA-rich biomass;
- (d) combining said PHA-reduced biomass with said PHA-rich biomass and returning the biomass combination to said culture of microorganisms to cause the culture to convert the carbon within the PHA-reduced biomass into PHA; and
- (e) collecting said PHA, including optionally separating said PHA from the PHA-reduced biomass.

20. The process of claim **19**, further comprising extracting said PHA from said removed portion of said microorganism-derived biomass comprising PHA is accomplished by i) mixing said removed portion with an extraction agent comprising solvents, enzymes, surfactants, acids, bases, hypochlorite, peroxides, polymers, bleaches, ozone, EDTA or combinations thereof or, ii) subjecting said removed portion of said microorganism-derived biomass comprising PHA to a mechanism comprising solvent washing, chemical treat-

ment, microwave treatment, simple or fractional distillation, supercritical carbon dioxide, heat, or combinations thereof.

21. A method for producing PHA in a microorganism culture, comprising the steps of:

- (a) providing a culture of microorganisms comprising microorganism-derived biomass, said microorganism-derived biomass comprising PHA, wherein said culture of microorganisms can metabolize both a carbon containing gas and carbon within a PHA-reduced biomass to generate PHA;
- (b) removing a portion of said microorganism-derived biomass comprising PHA from said culture of microorganisms and subjecting said microorganism-derived biomass comprising PHA to a PHA removal process, wherein said removal process results in (i) isolated PHA and (ii) PHA-reduced biomass having at least 10% less PHA content as compared to a microorganism-derived biomass not subjected to the removal, wherein said PHA-reduced biomass comprises carbon;
- (c) purifying the isolated PHA; and
- (d) combining said PHA-reduced biomass with said PHA-rich biomass and returning the biomass combination to said culture of microorganisms to cause the culture to convert the carbon within the PHA-reduced biomass into PHA.

22. The process of claim **21**, further comprising a step of introducing a light during said growth period.

23. The process of claim **22**, wherein said light is introduced using light emitting device to influence a metabolism of said culture.

24. The process of claim **22**, wherein said light is emitted by a device that is charged by inserting two leads of a 115V AC power source into said culture medium.

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